**Inter Pipeline Propylene Ltd.**
**Propane Dehydrogenation Project**

# PURLIN AND GIRT SIZING CALCULATION
# PDH-00-ST-CAL-0008

This document has been revised as indicated below.  Please destroy all previous revisions.

| Rev | Date | Revision Description | PREP | CHKD | APP'D | MANAGEMENT | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fluor | IPPL |
| 0 | 07-09-2017 | Issued for Information – Project 1202 | AKV | LT | | | |
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**PURLIN AND GIRT SIZING CALCULATION**

**Project:** Propane Dehydrogenation Project
**Location:** Strathcona County, AB
**Title:** Purlin and Girt Sizing Calculation
**Calculation No.:** PDH-00ST-CAL-0008
**By:** AKV

**Contract No.:** A6KV
**Date:** 07-Sep-2017

**Check Level:** 2B
**Chk'd:** LT

**Design Specifications:**
Structural Design Criteria: See specification PDH-00-ST-SPC-0001 Structural Engineering Design Criteria

**Approvals for First Issue <<*typically Issued for Design*>>**

| | Name (Print) | Signature | Date (yr/mo/day) |
|---|---|---|---|
| Originator | Ana Kharla Valerio | | 2017-Sep-07 |
| Checker | Lyndon Trombley | Digitally signed by tro82955<br>DN: cn=tro82955, email=Lyndon.Trombley@fluor.com<br>Date: 2017.11.28 14:27:52 -07'00' | 2017-Sep-07 |
| Engineer of Record | Ana Kharla Valerio | | 2017-Sep-07 |

1.1 Record of Revisions

| Rev. No. | Reason for Revision (Only for changes after first issue) | Date (yr/mo/day) | Engineer | Checked By |
|---|---|---|---|---|
| | | | | |

**PURLIN AND GIRT SIZING CALCULATION**

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Appendices

**PURLIN AND GIRT SIZING CALCULATION**

# 3. A UNCERTAINTY AND RISK (SEE JDB 6.99 ATTACHMENT 04)

| Uncertainty | Mitigation (Design Requirement) | Remaining Risk | Comments | Resolution of Uncertainty |
|---|---|---|---|---|
| Wall and roof material to be used not identified. | Assumed weights for wall and roof dead load are based on the governing dead load weight between 3 manufacturers (AWIP, Kingspan, Metlspan) specified on Architectural Specification (PDH-00AR-SPC-0001-10) | No remaining risk. | | |

# 4. COMPUTER FILE INDEX

The purpose of this form is to keep track of “calculation computer files” and to assign filenames using a standard format. This will assist in organizing data at a later date. All calculation files shall be stored under \215E_014_R_Proj_Calculations_Sketches\Calculations\PDH-00-ST-CAL-0008 and the respective subdirectory for the calculation folder in accordance to the Job Data Book.

| File Name | Description | Directory Location |
|---|---|---|
| PDH-00-ST-CAL-0008_rev0.pdf | Purlin and Girt Sizing Calculation | W:\215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_014_R_Proj_Calculations_Sketches\PDH-00-ST-CAL-0008_0000-SS-Purlin and Girt Sizing Calculation\Calculation Files |
| PDH-00-ST-CAL-0008_001.docx | Calculation Report Template | |
| PDH-00-ST-CAL-0008_002.xmcd | Girt Design for Building – 6m span | |
| PDH-00-ST-CAL-0008_003.xmcd | Girt Design for Building – 7m span | |
| PDH-00-ST-CAL-0008_004.xmcd | Purlin Design for Building | |
| PDH-00-ST-CAL-0008_005.xmcd | Girt Design for Stair Tower Module | |
| PDH-00-ST-CAL-0008_006.xmcd | Purlin Design for Stair Tower Module | |
| PDH-00-ST-CAL-0008_007.xmcd | Snow Load Calculation for Stair Tower | |
| PDH-00-ST-CAL-0008_008.xmcd | Roof/Wall Panel Check for Building | |

**PURLIN AND GIRT SIZING CALCULATION**

| | | |
|---|---|---|
| PDH-00-ST-CAL-0008_009.xmcd | Roof/Wall Panel Check for Stair Tower | W:\215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_014_R_Proj_Calculations_Sketches\PDH-00-ST-CAL-0008_0000-SS-Purlin and Girt Sizing Calculation\Calculation Files |
| PDH-00-ST-CAL-0008_010.xmcd | Design of Steel frame for openings | |
| PDH-00-ST-CAL-0008_011.r3d | RISA 3D model for design of steel frame for openings | |
| PDH-00-ST-CAL-0008_012.xmcd | Roof/Wall Panel Check for Stair Tower – 1m spacing | |
| PDH-00-ST-CAL-0008_013.xmcd | C310x31 Mrx Calculation | |
| PDH-00-ST-CAL-0008_014.xmcd | Check for Channel at Roof Perimeter – Stair Tower | |
| PDH-00-ST-CAL-0008_015.xmcd | Check for Channel at Roof Perimeter – Building | |

# 5. DESIGN CRITERIA

## 1.1 General

The loads used in this calculation are in kilonewtons (kN).
The dimensions are in meter and/or millimeter (m, mm).
Structural elements will be treated as a dead load.

## 1.2 Design Codes

### 1.2.1 Codes and Standards

| | |
|---|---|
| NBCC 2010 | National Building Code of Canada 2010 |
| ABC 2014 | Alberta Building Code 2014 |
| CAN/CSA-S16 | Limit States Design |
| CSA A23.3 | Design of Concrete Structures |

### 1.2.2 Specifications and Recommended Practices

| | |
|---|---|
| PDH-00-ST-SPC-0001 | Structural Engineering Design Criteria |
| PDH-00-ST-SPC-0005 | Fabrication and Erection of Structural Steel |
| PDH-00-ST-SPC-0006 | Concrete Materials |
| PDH-00-ST-SPC-0007 | Concrete Construction |
| PDH-00-ST-SPC-0008 | Concrete Reinforcing |
| PDH-00-ST-SPC-0012 | Driven Steel Piles |

## 1.3 Reference and Documentation

Job Data Book Section 2 — Structural Engineering Instructions

## 1.4 Computer Programs

RISA 3D 14.0.1 — Mathcad 15.0

## 1.5 Materials

| | |
|---|---|
| W Sections | CSA G40.21, Grade 350W/ASTM A992 |
| Channels | CSA G40.21, Grade 300W |

# 6. DESIGN SUMMARY

This calculation covers standardized design of purlin and girt for the buildings in the ISBL area and the modular stair tower in 01RE/02HP area. It also includes design of steel frame section for the openings in the buildings.

Design dead load assumptions for purlin and girt design are based on the governing roof and wall material of the 3 manufacturers (i.e. AWIP, Kingspan, Metlspan) specified on architectural specification PDH-00AR-SPC-0001-10.

Purlin and girt are designed based on 6m span on stair tower module and 7m span on building.

7. Summary of Member Sizes to be used:
   Building: Purlin (6m and 7m span) - W310x39
   Girt (6m span) - W200x36
   Girt (7m span) - W200x46
   Eave Girt (6m and 7m span) - C310x31
   Stair Tower: Purlin - W200x36
   Girt - W200x36
   Eave Girt - C250x30
   Openings: with 0-2 girt interruptions - C200x17
   with 3 or more girt interruptions - C310x31

8. Snow Load Calculation:

a) Snow load on Main Roof:

| | |
|---|---|
| $S_s := 1.6\text{kPa}$ | Ground Snow Load (3.2.11 of PDH-00-ST-SPC-0001) |
| $S_r := 0.1\text{kPa}$ | Associated Rain Load (3.2.11 of PDH-00-ST-SPC-0001) |
| $C_w := 1$ | Wind Exposure Factor (Cl 4.1.6.2 NBCC 2005) |
| $C_s := 1$ | Slope Factor (Cl 4.1.6.2 NBCC 2005) |
| $C_a := 1$ | Shape Factor (Cl 4.1.6.2 NBCC 2005) |
| $C_b := 0.8$ | Basic Roof Snow Load Factor (Cl 4.1.6.2 NBCC 2005) |
| $I_{S.ULS} := 1.15$ | Importance Factor (3.2.11 of PDH-00-ST-SPC-0001) |
| $I_{S.SLS} := 0.9$ | Importance Factor (3.2.11 of PDH-00-ST-SPC-0001) |

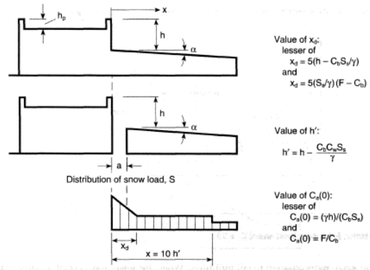
$S_{ULS} := I_{S.ULS} \cdot \left[\left[S_s \cdot \left(C_b \cdot C_w \cdot C_s \cdot C_a\right)\right] + S_r\right] = 1.59 \cdot \text{kPa}$ Snow Load ULS

$S_{SLS} := I_{S.SLS} \cdot \left[\left[S_s \cdot \left(C_b \cdot C_w \cdot C_s \cdot C_a\right)\right] + S_r\right] = 1.24 \cdot \text{kPa}$ Snow Load SLS

b) Snow Load accumulation on Stair Tower Roof: for HPC Building (Governs)

Calculation of Snow Drift load based on Fig. G-5

## PURLIN AND GIRT SIZING CALCULATION




| x | Factors[2] | | |
|---|---|---|---|
| | $C_w$ | $C_s$ | $C_a$ |
| 0 | 1.0 | $f(\alpha)$[1] | $C_a(0)$ |
| $0 < x \le x_d$ | 1.0 | $f(\alpha)$[1] | $C_a(0) - \frac{(C_a(0) - C_a(x_d))\,x}{x_d}$ |
| $x_d < x \le 10\,h'$ | 1.0 | $f(\alpha)$[1] | 1.0 |
| $> 10\,h'$ | 1.0[3] | $f(\alpha)$[1] | 1.0 |

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$h := 6.26\text{m}$ — height difference between upper and lower roof

$h_p := 0\text{m}$ — height of upper roof parapet

$w_u := 21.2\text{m}$ — smaller upper roof dimension

$l_u := 32.5\text{m}$ — larger upper roof dimension

$$l_c := 2w_u - \frac{w_u^{\,2}}{l_u} = 28.57\,\text{m}$$ Characteristic length of upper roof

$$\gamma_{snow} := 3\,\frac{\text{kN}}{\text{m}^3}$$ unit weight of snow (Commentary G, Paragraph 7)

$$h_{prime} := h - \frac{C_b \cdot C_w \cdot S_s}{\gamma_{snow}} = 5.83\,\text{m} \qquad 10 \cdot h_{prime} = 58.33\,\text{m}$$

$$F := \max\left[2, 0.35 \cdot \left[\frac{\gamma_{snow} \cdot l_c}{S_s} - 6 \cdot \left(\frac{\gamma_{snow} \cdot h_p}{S_s}\right)^2\right]^{0.5} + C_b\right] = 3.36$$

$$x_d := \min\left[5 \cdot \left(h - \frac{C_b \cdot S_s}{\gamma_{snow}}\right), 5 \cdot \left(\frac{S_s}{\gamma_{snow}}\right) \cdot \left(F - C_b\right)\right] = 6.83\,\text{m}$$ length of drift

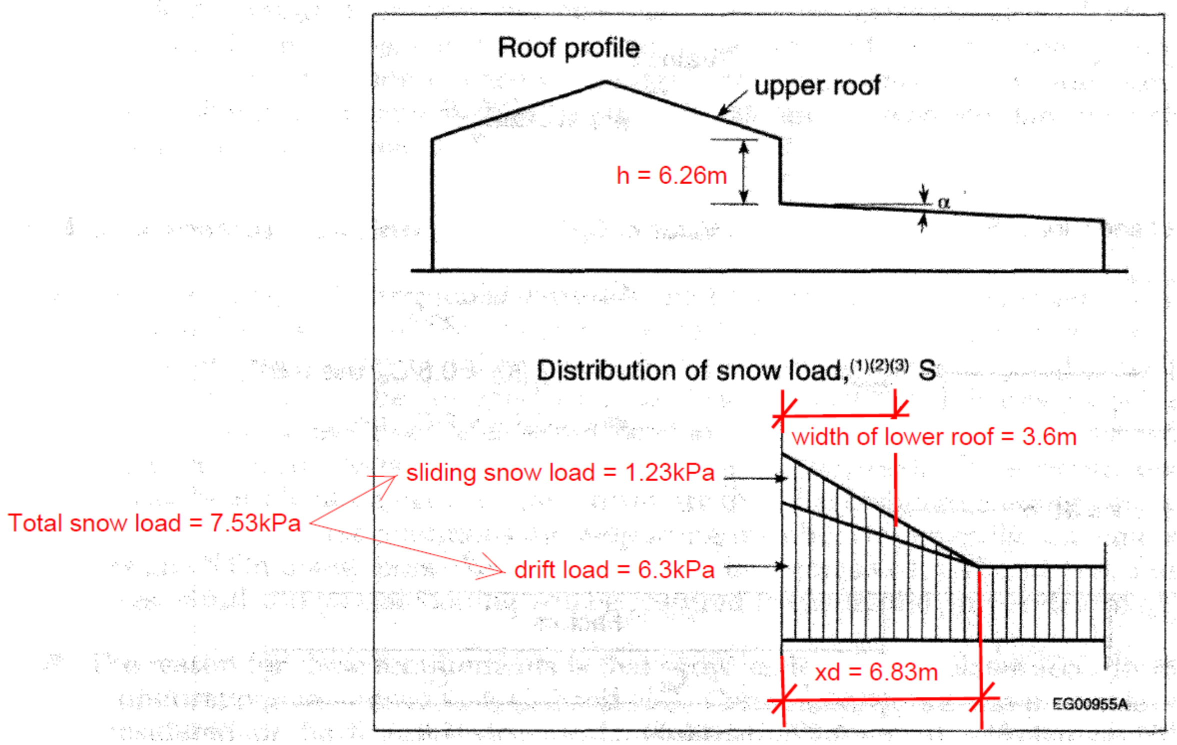
$$C_{a.0} := \min\left(\frac{\gamma_{snow} \cdot h}{C_b \cdot S_s}, \frac{F}{C_b}\right) = 4.2$$ Ca factore at edge of lower roof adjacent to building

$$S_{drift} := I_{S.ULS} \cdot \left[\left[S_s \cdot \left(C_b \cdot C_w \cdot C_s \cdot C_{a.0}\right)\right] + S_r\right] = 6.3 \cdot \text{kPa}$$ Snow Drift Load on Lower roof

Calculation of Sliding Snow Load based on Fig. G-7 and paragraph 41

As per NBCC 2010 Commentary G paragraph 41, sliding snow load is 50% of Case 1 snow (1.59kPa) and can be distributed based on the relative sizes of the roof. The distribution of the sliding snow load is based on the assumption that the load from one end of the upper roof (i.e. horizontal length from the ridge to the low eave) will slide at the lower roof at distance xd. This is shown in Fig. G-7

$$S_{slide} := \frac{0.5 \cdot S_{ULS} \cdot \frac{w_u}{2}}{x_d} = 1.23 \cdot kPa$$ Sliding Snow Load




$$S_{lower} := S_{drift} + S_{slide} = 7.53 \cdot kPa$$ total snow load on lower roof adjacent to building wall

$$h_{snow} := \frac{S_{lower}}{\gamma_{snow}} = 2.51\,m$$ equivalent height of snow

Since the height between the upper and lower roof (6.26m) is greater than the height of snow (2.51m) for the total drift and sliding load, no reduction of snow load is allowed.

**PURLIN AND GIRT SIZING CALCULATION**

Snow load at low eave of stair tower roof:

$w_l := 3.6m$ width of lower roof

$l_l := 6.8m$ length of lower roof

$$S_{eave} := \frac{(x_d - w_l) \cdot (S_{lower} - S_{ULS})}{x_d} + S_{ULS} = 4.4 \cdot kPa$$

Snow load at stair tower rafter: adjacent to building wall- $S_{lower} \cdot \frac{l_l}{2} = 25.61 \cdot \frac{kN}{m}$

eave - $S_{eave} \cdot \frac{l_l}{2} = 14.96 \cdot \frac{kN}{m}$

## PURLIN AND GIRT SIZING CALCULATION

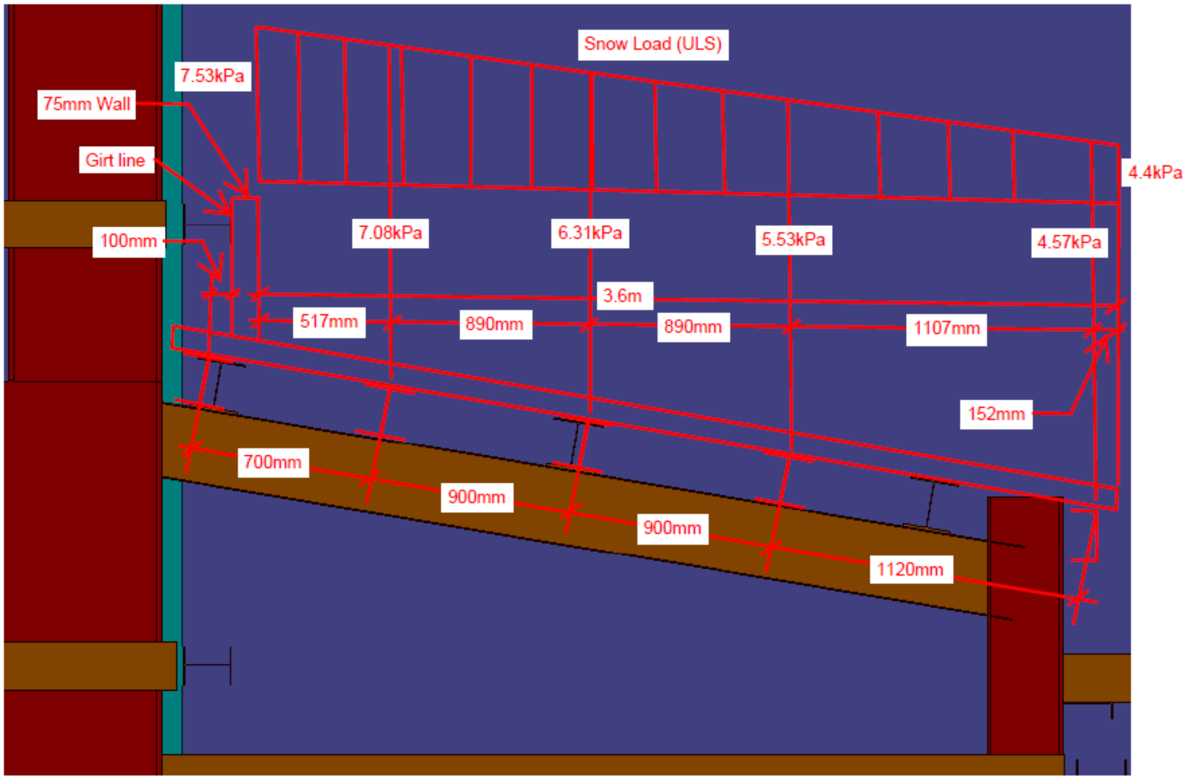
Snow load for Purlin Design:

Upper Roof - $S_{upper} := 1.59kPa$ Snow load for Upper Roof (ULS)

Lower Roof - $S_{lower} = 7.53 \cdot kPa$ Snow load for Lower Roof adjacent to building wall(ULS)

$S_{eave} = 4.4 \cdot kPa$ Snow load for Lower Roof eave (ULS)

Purlin spacing as shown below:




Critical purlin location is at 517mm from face of wall

Snow load at critical purlin location:

$$S_{cr.purlin.ULS} := \frac{(x_d - 0.517m) \cdot (S_{lower} - S_{ULS})}{x_d} + S_{ULS} = 7.08 \cdot kPa$$

$$S_{cr.purlin.SLS} := \frac{S_{cr.purlin.ULS}}{1.15} \cdot 0.9 = 5.54 \cdot kPa$$

**PURLIN AND GIRT SIZING CALCULATION**

# 9. Purlin and Girt Design for Building

## 9.1 Purlin Design: HPC Building

Purlin Section - W310x39 Purlin Spacing - $s := 1.5\text{m}$ Purlin Span - $L := 7\text{m}$

Section Properties:

$wt := 0.38\frac{\text{kN}}{\text{m}}$ section dead load

$b := 165\text{mm}$ flange width

$d := 310\text{mm}$ depth

$t_w := 5.8\text{mm}$ web thickness

$t_f := 9.7\text{mm}$ flange thickness

$I_x := 85.1\cdot10^6\text{mm}^4$ $I_y := 7.27\cdot10^6\text{mm}^4$ $J := 126\cdot10^3\cdot\text{mm}^4$

$S_x := 549\cdot10^3\text{mm}^3$ $S_y := 88.1\cdot10^3\text{mm}^3$ $C_w := 164\cdot10^9\cdot\text{mm}^6$

$Z_x := 610\cdot10^3\text{mm}^3$ $Z_y := 135\cdot10^3\text{mm}^3$

$F_y := 350\text{MPa}$ $E_s := 200\text{GPa}$ $G_s := 77\text{GPa}$ $\phi_s := 0.9$

Roof Details: $\theta := 9.462\text{deg}$

4" Kingzip Standing Seam Panels 24 Ga (governing panel) - as per PDH-00-AR-SPC-0001-10

$wt_{panel} := 3.05\text{psf} = 0.146\cdot\text{kPa}$

Loads: $q_{live} := 1\text{kPa}$ Live load

$q_e := 1\text{kPa}$ Equipment/Electrical Load

$q_{snow} := 1.59\text{kPa}$ Snow Load (ULS) - refer to page 8

Wind Load Calculation for Purlins:

$I_{ULS} := 1.15$ $I_{SLS} := 0.75$ Importance Factor

$q := 0.43\text{kPa}$ reference velocity pressure

$El_{roof} := 127.3\text{m}$ roof elevation

$El_{grade} := 100\text{m}$ grade elevation

$$C_e := \left(\frac{El_{roof} - El_{grade}}{10\text{m}}\right)^{0.2} = 1.22$$ exposure factor

$C_g := 2.5$ external gust effect factor for small elements incl. cladding

$C_p := 2.3$ external suction coeefficient for roofing and cladding

**PURLIN AND GIRT SIZING CALCULATION**

$C_{gi} := 2$ internal gust effect factor

$C_{pi.s} := 0.45$ internal suction coefficient

$C_{pi.p} := 0.3$ internal presure coefficient

$WL_{external.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_p = 3.476 \cdot kPa$ for External Suction

$WL_{internal.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.54 \cdot kPa$ for Internal Suction

$WL_{internal.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.36 \cdot kPa$ for Internal Pressure

$WL_{external.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_p = 2.27 \cdot kPa$ for External Suction

$WL_{internal.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.35 \cdot kPa$ for Internal Suction

$WL_{internal.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.24 \cdot kPa$ for Internal Pressure

Total Wind Load:

Downward Force - $WL_{ULS.down} := WL_{internal.s.ULS} = 0.54 \cdot kPa$

$WL_{SLS.down} := WL_{internal.s.SLS} = 0.35 \cdot kPa$

Upward Force - $WL_{ULS.up} := WL_{external.ULS} + WL_{internal.p.ULS} = 3.84 \cdot kPa$

$WL_{SLS.up} := WL_{external.SLS} + WL_{internal.p.SLS} = 2.5 \cdot kPa$

Factored Linear load on purlins

<u>Load Combination 1: 1.25DL + 1.5SL +0.5LL</u>

$$L_{y1.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta) + 0.5 \cdot q_{live} \cdot s\right] \cdot \cos(\theta) = 6.81 \cdot \frac{kN}{m}$$

$$L_{x1.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta) + 0.5 \cdot q_{live} \cdot s\right] \cdot \sin(\theta) = 1.13 \cdot \frac{kN}{m}$$

<u>Load Combination 2: 1.25DL + 1.5SL + 0.4WLdown</u>

$$L_{y2.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \cos(\theta) + 0.4 \cdot WL_{ULS.down} \cdot s = 6.4 \cdot \frac{kN}{m}$$

$$L_{x2.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \sin(\theta) = 1.01 \cdot \frac{kN}{m}$$

<u>Load Combination 3: 1.25DL + 1.4WLdown + 0.5SL</u>

$$L_{y3.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 0.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \cos(\theta) + 1.4 \cdot WL_{ULS.down} \cdot s = 4.891 \cdot \frac{kN}{m}$$

$$L_{x3.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 0.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \sin(\theta) = 0.625 \cdot \frac{kN}{m}$$

<u>Load Combination 4: 1.25DL - 1.4WLup - does not include equipment weight which is conservative</u>

$$L_{y4.ULS} := 1.25 \cdot \left(wt + wt_{panel} \cdot s\right) \cdot \cos(\theta) - 1.4 \cdot WL_{ULS.up} \cdot s = -7.322 \cdot \frac{kN}{m}$$

$$L_{x4.ULS} := 1.25 \cdot \left(wt + wt_{panel} \cdot s\right) \sin(\theta) = 0.123 \cdot \frac{kN}{m}$$

**PURLIN AND GIRT SIZING CALCULATION**

Governing Factored Load: $y_{ULS} := \left(L_{y1.ULS} \;\; L_{y2.ULS} \;\; L_{y3.ULS} \;\; L_{y4.ULS}\right)$

$x_{ULS} := \left(L_{x1.ULS} \;\; L_{x2.ULS} \;\; L_{x3.ULS} \;\; L_{x4.ULS}\right)$

$L_{y.ULS.down} := \max\left(y_{ULS}\right) = 6.81 \cdot \frac{kN}{m}$ $\qquad L_{x.ULS} := \max\left(x_{ULS}\right) = 1.13 \cdot \frac{kN}{m}$

$L_{y.ULS.up} := \min\left(y_{ULS}\right) = -7.32 \cdot \frac{kN}{m}$

Service Linear Load on purlins

Load Combination 1: DL + SL +LL

$$L_{y1.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta) + q_{live} \cdot s\right] \cdot \cos(\theta) = 5.37 \cdot \frac{kN}{m}$$

$$L_{x1.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta) + q_{live} \cdot s\right] \cdot \sin(\theta) = 0.89 \cdot \frac{kN}{m}$$

Load Combination 2: DL + SL + WLdown

$$L_{y2.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \cos(\theta) + 0.4 \cdot WL_{SLS.down} \cdot s = 4.1 \cdot \frac{kN}{m}$$

$$L_{x2.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \sin(\theta) = 0.65 \cdot \frac{kN}{m}$$

Load Combination 3: DL - WLup - does not include equipment weight which is conservative

$$L_{y3.SLS} := \left(wt + wt_{panel} \cdot s\right) \cdot \cos(\theta) - WL_{SLS.up} \cdot s = -3.164 \cdot \frac{kN}{m}$$

$$L_{x3.SLS} := \left(wt + wt_{panel} \cdot s\right) \cdot \sin(\theta) = 0.098 \cdot \frac{kN}{m}$$

Governing Service Load: $y_{SLS} := \left(L_{y1.SLS} \;\; L_{y2.SLS} \;\; L_{y3.SLS}\right)$

$x_{SLS} := \left(L_{x1.SLS} \;\; L_{x2.SLS} \;\; L_{x3.SLS}\right)$

$$L_{y.SLS.down} := \max\left(y_{SLS}\right) = 5.37 \cdot \frac{kN}{m}$$

$$L_{y.SLS.up} := \min\left(y_{SLS}\right) = -3.16 \cdot \frac{kN}{m}$$

$$L_{x.SLS} := \max\left(x_{SLS}\right) = 0.89 \cdot \frac{kN}{m}$$

## PURLIN AND GIRT SIZING CALCULATION

### Purlin Design

Section Class

$$\text{Flange}_{class} := \begin{cases} 1 & \text{if } \dfrac{b}{2\cdot t_f} \le \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{200}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 2$$

$$\text{Web}_{class} := \begin{cases} 1 & \text{if } \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1900}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 1$$

$$\text{Section}_{class} := \max\left(\text{Flange}_{class}, \text{Web}_{class}\right) = 2$$

Check Bending Capacity

*For laterally supported members:* This is applicable to purlins when force direction is downward. During this case the top flange is in compression and the roof panel provides lateral support to the compression flange.
For moment capacity in weak direction: Since load is applied on top flange, only half the section capacity is considered for design.

$$M_{fx.down} := \left| \frac{L_{y.ULS.down}\cdot L^2}{8} \right| = 41.7\cdot kN\cdot m$$

$$M_{fy.down} := \left| \frac{L_{x.ULS}\cdot L^2}{8} \right| = 6.95\cdot kN\cdot m$$

$$M_{rx.lat} := \begin{cases} \phi_s\cdot Z_x\cdot F_y & \text{if } \text{Section}_{class} = 1 \vee 2 \\ \phi_s\cdot S_x\cdot F_y & \text{if } \text{Section}_{class} = 3 \\ \text{"use Class 1,2 or 3"} & \text{otherwise} \end{cases} = 192.15\cdot kN\cdot m$$

$$M_{ry} := \frac{1}{2}\cdot\left(\phi_s\cdot S_y\cdot F_y\right) = 13.88\cdot kN\cdot m$$

$$UC_{latertally.supported} := \frac{M_{fx.down}}{M_{rx.lat}} + \frac{M_{fy.down}}{M_{ry}} = 0.718$$ less than 1 OK

**PURLIN AND GIRT SIZING CALCULATION**

*For laterally unsupported members:* This is applicable to purlins when force direction is upward. During this case the bottom flange is in compression and no lateral restraint is provided to prevent lateral buckling.
For moment capacity in weak direction: Since load is applied on top flange, only half the section capacity is considered for design.

$$M_{fx.up} := \left| \frac{L_{y.ULS.up} \cdot L^2}{8} \right| = 44.85 \cdot kN \cdot m$$

$$M_{fy.up} := \left| \frac{\text{lookup}\left(L_{y.ULS.up}, y_{ULS}, x_{ULS}\right) \cdot L^2}{8} \right| = 0.75 \cdot kN \cdot m$$

$$M_{max} := M_{fx.up} = 44.85 \cdot kN \cdot m$$

$$M_a := \left| L_{y.ULS.up} \cdot \frac{3L^2}{32} \right| = 33.637 \cdot kN \cdot m$$

factored bending moment at 1/4 point of unbraced segment

$$M_b := M_{fx.up} = 44.85 \cdot kN \cdot m$$

factored bending moment at midpoint of unbraced segment

$$M_c := \left| L_{y.ULS.up} \cdot \frac{3L^2}{32} \right| = 33.637 \cdot kN \cdot m$$

factored bending moment at 3/4 point of unbraced segment

$$\omega_2 := \min\left(2.5, \frac{4 \cdot M_{max}}{\sqrt{M_{max}^2 + 4 \cdot M_a^2 + 7 \cdot M_b^2 + 4 \cdot M_c^2}}\right) = 1.131$$

$$M_u := \frac{\omega_2 \cdot \pi}{L} \cdot \sqrt{E_s \cdot I_y \cdot G_s \cdot J + \left(\frac{\pi \cdot E_s}{L}\right)^2 \cdot I_y \cdot C_w} = 78.19 \cdot kN \cdot m$$

critical elastic moment

$$M_{rx.nolat} := \begin{cases} \text{if } Section_{class} = 1 \vee 2 \\ \quad \begin{cases} 1.15 \cdot \phi_s \cdot Z_x \cdot F_y \cdot \left(1 - \frac{0.28 \cdot Z_x \cdot F_y}{M_u}\right) & \text{if } M_u > 0.67 \cdot Z_x \cdot F_y \\ \phi_s \cdot M_u & \text{otherwise} \end{cases} \\ \text{if } Section_{class} = 3 \\ \quad \begin{cases} 1.15 \cdot \phi_s \cdot S_x \cdot F_y \cdot \left(1 - \frac{0.28 \cdot S_x \cdot F_y}{M_u}\right) & \text{if } M_u > 0.67 \cdot S_x \cdot F_y \\ \phi_s \cdot M_u & \text{otherwise} \end{cases} \end{cases} = 70.37 \cdot kN \cdot m$$

**PURLIN AND GIRT SIZING CALCULATION**

$$UC_{nolateral.support} := \frac{M_{fx.up}}{M_{rx.nolat}} + \frac{M_{fy.up}}{M_{ry}} = 0.692$$ less than 1 OK

Check Deflection

$$\Delta_y := \frac{5 \cdot \max\left(L_{y.SLS.down}, L_{y.SLS.up}\right) \cdot (L)^4}{384 \cdot E_s \cdot I_x} = 9.86 \cdot mm$$

$$\Delta_x := \frac{5 \cdot L_{x.SLS} \cdot (L - 0.8m)^4}{384 \cdot E_s \cdot I_y} = 11.83 \cdot mm$$

$$\Delta_{allowable} := \frac{L}{240} = 29.17 \cdot mm$$ greater than x,y deflection OK

**PURLIN AND GIRT SIZING CALCULATION**

## 9.2 For Girt Span at 6m

Girt Design: Builidng (for 6m span)

GirtSection - W200x36 Girt Spacing - $s := 1.8m$ Girt Span - $L := 6m$

Section Properties:

$wt := 0.352\frac{kN}{m}$ section dead load

$b := 165mm$ flange width

$d := 201mm$ depth

$t_w := 6.2mm$ web thickness

$t_f := 10.2mm$ flange thickness

$I_x := 34.4 \cdot 10^6 mm^4$ $I_y := 7.64 \cdot 10^6 mm^4$ $J := 145 \cdot 10^3 \cdot mm^4$

$S_x := 342 \cdot 10^3 mm^3$ $S_y := 92.6 \cdot 10^3 mm^3$ $C_w := 69.6 \cdot 10^9 \cdot mm^6$

$Z_x := 379 \cdot 10^3 mm^3$ $Z_y := 141 \cdot 10^3 mm^3$

$F_y := 350MPa$ $E_s := 200GPa$ $G_s := 77GPa$ $\phi_s := 0.9$

4.5" AWIP fiRe (governing panel) - as per PDH-00-AR-SPC-0001-10

$wt_{panel} := 5.5psf = 0.263kPa$

Loads: Wind Load Calculation for Girts:

$I_{ULS} := 1.15$ $I_{SLS} := 0.75$ Importance Factor

$q := 0.43kPa$ reference velocity pressure

$El_{roof} := 128m$ roof elevation

$El_{grade} := 100m$ grade elevation

$$C_e := \left(\frac{El_{roof} - El_{grade}}{10m}\right)^{0.2} = 1.23$$ exposure factor

$C_g := 2.5$ external gust effect factor for small elements incl. cladding

$C_{p.s} := 1.2$ external suction coefficient for cladding

$C_{p.p} := 0.9$ external pressure coefficient for cladding

$C_{gi} := 2$ internal gust effect factor

$C_{pi.s} := 0.45$ internal suction coefficient

## PURLIN AND GIRT SIZING CALCULATION

$C_{pi.p} := 0.3$ internal presure coefficient

ULS Load:

$WL_{external.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.s} = 1.823 \cdot kPa$ for External Suction

$WL_{external.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.p} = 1.37 \cdot kPa$ for External Pressure

$WL_{internal.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.55 \cdot kPa$ for Internal Suction

$WL_{internal.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.36 \cdot kPa$ for Internal Pressure

SLS Load:

$WL_{external.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.s} = 1.189 \cdot kPa$ for External Suction

$WL_{external.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.p} = 0.89 \cdot kPa$ for External Pressure

$WL_{internal.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.357 \cdot kPa$ for Internal Suction

$WL_{internal.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.24 \cdot kPa$ for Internal Pressure

Total Wind Load:

Wall Suction (Negative WL) - $WL_{suction.ULS} := WL_{external.s.ULS} + WL_{internal.p.ULS} = 2.19 \cdot kPa$

$WL_{suction.SLS} := WL_{external.s.SLS} + WL_{internal.p.SLS} = 1.43 \cdot kPa$

Wall Pressure (Positive WL) - $WL_{pressure.ULS} := WL_{external.p.ULS} + WL_{internal.s.ULS} = 1.91 \cdot kPa$

$WL_{pressure.SLS} := WL_{external.p.SLS} + WL_{internal.s.SLS} = 1.25 \cdot kPa$

$q_e := 0.5kPa$ additional load due to equipment/tray/pipe supported on girts

Factored Linear load on girts

Load Combination: 1.25DL + 1.4WL

$$L_{x.ULS} := 1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) = 2.16 \cdot \frac{kN}{m}$$

$$L_{y.suction.ULS} := 1.4 \cdot WL_{suction.ULS} \cdot s = 5.51 \cdot \frac{kN}{m}$$

$$L_{y.pressure.ULS} := 1.4 \cdot WL_{pressure.ULS} \cdot s = 4.82 \cdot \frac{kN}{m}$$

Service Linear Load on girts

Load Combination: DL + WL

$$L_{x.SLS} := \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) = 1.73 \cdot \frac{kN}{m}$$

$$L_{y.suction.SLS} := WL_{suction.SLS} \cdot s = 2.57 \cdot \frac{kN}{m}$$

$$L_{y.pressure.SLS} := WL_{pressure.SLS} \cdot s = 2.25 \cdot \frac{kN}{m}$$

**PURLIN AND GIRT SIZING CALCULATION**

**Girt Design**

Section Class

$$Flange_{class} := \begin{cases} 1 & \text{if } \dfrac{b}{2 \cdot t_f} \le \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2 \cdot t_f} \le \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2 \cdot t_f} \le \dfrac{200}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 2$$

$$Web_{class} := \begin{cases} 1 & \text{if } \dfrac{d - 2 \cdot t_f}{t_w} \le \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2 \cdot t_f}{t_w} \le \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2 \cdot t_f}{t_w} \le \dfrac{1900}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 1$$

$$Section_{class} := \max\left(Flange_{class}, Web_{class}\right) = 2$$

Check Bending Capacity

*For laterally supported members:* This is applicable to girts when flange with wall connection is in compression. During this case the flange is in compression and the wall panel provides lateral support to the flange.

For moment capacity in weak direction: Only half the section capacity is considered for design.

$$M_{fx.lat} := \left| \frac{L_{y.pressure.ULS} \cdot L^2}{8} \right| = 21.7 \cdot kN \cdot m$$

$$M_{fy.lat} := \left| \frac{L_{x.ULS} \cdot L^2}{8} \right| = 9.71 \cdot kN \cdot m$$

$$M_{rx.lat} := \begin{cases} \phi_s \cdot Z_x \cdot F_y & \text{if } Section_{class} = 1 \vee 2 \\ \phi_s \cdot S_x \cdot F_y & \text{if } Section_{class} = 3 \\ \text{"use Class 1,2 or 3"} & \text{otherwise} \end{cases} = 119.385 \cdot kN \cdot m$$

$$M_{ry} := \frac{1}{2} \cdot \left(\phi_s \cdot S_y \cdot F_y\right) = 14.58 \cdot kN \cdot m$$

$$UC_{lateral} := \frac{M_{fx.lat}}{M_{rx.lat}} + \frac{M_{fy.lat}}{M_{ry}} = 0.847$$ less than 1 OK

## PURLIN AND GIRT SIZING CALCULATION

*For laterally unsupported members:* This is applicable to girts when flange with wall connection is in tension During this case the flange is in tension and no lateral restraint is provided to prevent lateral buckling.
For moment capacity in weak direction: Only half the section capacity is considered for design.

$$M_{fx.nolat} := \left|\frac{L_{y.suction.ULS}\cdot L^2}{8}\right| = 24.8\cdot kN\cdot m$$

$$M_{fy.nolat} := \left|\frac{L_{x.ULS}\cdot L^2}{8}\right| = 9.71\cdot kN\cdot m$$

$$M_{max} := M_{fx.nolat} = 24.8\cdot kN\cdot m$$

$$M_a := \left|L_{y.suction.ULS}\cdot\frac{3L^2}{32}\right| = 18.6\cdot kN\cdot m$$

factored bending moment at 1/4 point of unbraced segment

$$M_b := M_{fx.nolat} = 24.8\cdot kN\cdot m$$

factored bending moment at midpoint of unbraced segment

$$M_c := \left|L_{y.suction.ULS}\cdot\frac{3L^2}{32}\right| = 18.6\cdot kN\cdot m$$

factored bending moment at 3/4 point of unbraced segment

$$\omega_2 := \min\left(2.5, \frac{4\cdot M_{max}}{\sqrt{M_{max}^2 + 4\cdot M_a^2 + 7\cdot M_b^2 + 4\cdot M_c^2}}\right) = 1.131$$

$$M_u := \frac{\omega_2\cdot\pi}{L}\cdot\sqrt{E_s\cdot I_y\cdot G_s\cdot J + \left(\frac{\pi\cdot E_s}{L}\right)^2\cdot I_y\cdot C_w} = 89.63\cdot kN\cdot m$$

critical elastic moment

$$M_{rx.nolat} := \begin{array}{l} \text{if } Section_{class} = 1 \vee 2 \\ \quad \left| \begin{array}{ll} 1.15\cdot\phi_s\cdot Z_x\cdot F_y\cdot\left(1 - \frac{0.28\cdot Z_x\cdot F_y}{M_u}\right) & \text{if } M_u > 0.67\cdot Z_x\cdot F_y \\ \phi_s\cdot M_u & \text{otherwise} \end{array} \right. \\ \text{if } Section_{class} = 3 \\ \quad \left| \begin{array}{ll} 1.15\cdot\phi_s\cdot S_x\cdot F_y\cdot\left(1 - \frac{0.28\cdot S_x\cdot F_y}{M_u}\right) & \text{if } M_u > 0.67\cdot S_x\cdot F_y \\ \phi_s\cdot M_u & \text{otherwise} \end{array} \right. \end{array} = 80.4\cdot kN\cdot m$$

## PURLIN AND GIRT SIZING CALCULATION

$$UC_{nolateral.support} := \frac{M_{fx.nolat}}{M_{rx.nolat}} + \frac{M_{fy.nolat}}{M_{ry}} = 0.974$$ less than 1 OK

Check Deflection

$$\Delta_y := \frac{5 \cdot \max\left(L_{y.pressure.SLS}, L_{y.suction.SLS}\right) \cdot (L)^4}{384 \cdot E_s \cdot I_x} = 6.3 \cdot mm$$

$$\Delta_x := \frac{5 \cdot L_{x.SLS} \cdot L^4}{384 \cdot E_s \cdot I_y} = 19.06 \cdot mm$$

$$\Delta_{allowable} := \frac{L}{180} = 33.33 \cdot mm$$ greater than x,y deflection OK

**PURLIN AND GIRT SIZING CALCULATION**

## 9.3 For Girt Span at 7m

Girt Design: Builidng (for 7m span)

GirtSection - W200x46 Girt Spacing - $s := 1.8m$ Girt Span - $L := 7m$

Section Properties:

$wt := 0.451 \frac{kN}{m}$ section dead load

$b := 203mm$ flange width

$d := 203mm$ depth

$t_w := 7.2mm$ web thickness

$t_f := 11mm$ flange thickness

$I_x := 45.4 \cdot 10^6 mm^4$ $I_y := 15.3 \cdot 10^6 mm^4$ $J := 220 \cdot 10^3 \cdot mm^4$

$S_x := 448 \cdot 10^3 mm^3$ $S_y := 151 \cdot 10^3 mm^3$ $C_w := 141 \cdot 10^9 \cdot mm^6$

$Z_x := 495 \cdot 10^3 mm^3$ $Z_y := 229 \cdot 10^3 mm^3$

$F_y := 350MPa$ $E_s := 200GPa$ $G_s := 77GPa$ $\phi_s := 0.9$

3" Panel Kingspan Shadowline KS42 24 Ga (governing panel) - as per PDH-00-AR-SPC-0001-10

$wt_{panel} := 2.86psf = 0.137 \cdot kPa$

Loads: Wind Load Calculation for Purlins:

$I_{ULS} := 1.15$ $I_{SLS} := 0.75$ Importance Factor

$q := 0.43kPa$ reference velocity pressure

$El_{roof} := 128m$ roof elevation

$El_{grade} := 100m$ grade elevation

$$C_e := \left(\frac{El_{roof} - El_{grade}}{10m}\right)^{0.2} = 1.23$$ exposure factor

$C_g := 2.5$ external gust effect factor for small elements incl. cladding

$C_{p.s} := 1.2$ external suction coefficient for cladding

$C_{p.p} := 0.9$ external pressure coefficient for cladding

$C_{gi} := 2$ internal gust effect factor

$C_{pi.s} := 0.45$ internal suction coefficient

**PURLIN AND GIRT SIZING CALCULATION**

$C_{pi.p} := 0.3$ internal presure coefficient

ULS Load:
$WL_{external.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.s} = 1.823 \cdot kPa$ for External Suction
$WL_{external.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.p} = 1.37 \cdot kPa$ for External Pressure
$WL_{internal.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.55 \cdot kPa$ for Internal Suction
$WL_{internal.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.36 \cdot kPa$ for Internal Pressure

SLS Load:
$WL_{external.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.s} = 1.189 \cdot kPa$ for External Suction
$WL_{external.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.p} = 0.89 \cdot kPa$ for External Pressure
$WL_{internal.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.357 \cdot kPa$ for Internal Suction
$WL_{internal.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.24 \cdot kPa$ for Internal Pressure

Total Wind Load:
Wall Suction (Negative WL) - $WL_{suction.ULS} := WL_{external.s.ULS} + WL_{internal.p.ULS} = 2.19 \cdot kPa$
$WL_{suction.SLS} := WL_{external.s.SLS} + WL_{internal.p.SLS} = 1.43 \cdot kPa$
Wall Pressure (Positive WL) - $WL_{pressure.ULS} := WL_{external.p.ULS} + WL_{internal.s.ULS} = 1.91 \cdot kPa$
$WL_{pressure.SLS} := WL_{external.p.SLS} + WL_{internal.s.SLS} = 1.25 \cdot kPa$

$q_e := 0.5kPa$ additional load due to equipment/tray/pipe supported on girts

Factored Linear load on girts

Load Combination: 1.25DL + 1.4WL

$$L_{x.ULS} := 1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) = 2 \cdot \frac{kN}{m}$$

$$L_{y.suction.ULS} := 1.4 \cdot WL_{suction.ULS} \cdot s = 5.51 \cdot \frac{kN}{m}$$

$$L_{y.pressure.ULS} := 1.4 \cdot WL_{pressure.ULS} \cdot s = 4.82 \cdot \frac{kN}{m}$$

Service Linear Load on girts

Load Combination: DL + WL

$$L_{x.SLS} := \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) = 1.6 \cdot \frac{kN}{m}$$

$$L_{y.suction.SLS} := WL_{suction.SLS} \cdot s = 2.57 \cdot \frac{kN}{m}$$

$$L_{y.pressure.SLS} := WL_{pressure.SLS} \cdot s = 2.25 \cdot \frac{kN}{m}$$

## PURLIN AND GIRT SIZING CALCULATION

### Girt Design

Section Class

$$Flange_{class} := \begin{cases} 1 & \text{if } \dfrac{b}{2\cdot t_f} \le \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{200}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 3$$

$$Web_{class} := \begin{cases} 1 & \text{if } \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1900}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 1$$

$$Section_{class} := \max(Flange_{class}, Web_{class}) = 3$$

Check Bending Capacity

*For laterally supported members:* This is applicable to girts when flange with wall connection is in compression. During this case the flange is in compression and the wall panel provides lateral support to the flange.
For moment capacity in weak direction: Only half the section capacity is considered for design.

$$M_{fx.lat} := \left| \frac{L_{y.pressure.ULS}\cdot L^2}{8} \right| = 29.54\cdot kN\cdot m$$

$$M_{fy.lat} := \left| \frac{L_{x.ULS}\cdot L^2}{8} \right| = 12.23\cdot kN\cdot m$$

$$M_{rx.lat} := \begin{cases} \phi_s\cdot Z_x\cdot F_y & \text{if } Section_{class} = 1 \vee 2 \\ \phi_s\cdot S_x\cdot F_y & \text{if } Section_{class} = 3 \\ \text{"use Class 1,2 or 3"} & \text{otherwise} \end{cases} = 141.12\cdot kN\cdot m$$

$$M_{ry} := \frac{1}{2}\cdot(\phi_s\cdot S_y\cdot F_y) = 23.78\cdot kN\cdot m$$

$$UC_{lateral} := \frac{M_{fx.lat}}{M_{rx.lat}} + \frac{M_{fy.lat}}{M_{ry}} = 0.724$$ less than 1 OK

## PURLIN AND GIRT SIZING CALCULATION

*For laterally unsupported members:* This is applicable to girts when flange with wall connection is in tension During this case the flange is in tension and no lateral restraint is provided to prevent lateral buckling.
For moment capacity in weak direction: Only half the section capacity is considered for design.

$$M_{fx.nolat} := \left| \frac{L_{y.suction.ULS} \cdot L^2}{8} \right| = 33.76 \cdot kN \cdot m$$

$$M_{fy.nolat} := \left| \frac{L_{x.ULS} \cdot L^2}{8} \right| = 12.23 \cdot kN \cdot m$$

$$M_{max} := M_{fx.nolat} = 33.76 \cdot kN \cdot m$$

$$M_a := \left| L_{y.suction.ULS} \cdot \frac{3L^2}{32} \right| = 25.3 \cdot kN \cdot m$$

factored bending moment at 1/4 point of unbraced segment

$$M_b := M_{fx.nolat} = 33.76 \cdot kN \cdot m$$

factored bending moment at midpoint of unbraced segment

$$M_c := \left| L_{y.suction.ULS} \cdot \frac{3L^2}{32} \right| = 25.32 \cdot kN \cdot m$$

factored bending moment at 3/4 point of unbraced segment

$$\omega_2 := \min\left(2.5, \frac{4 \cdot M_{max}}{\sqrt{M_{max}^2 + 4 \cdot M_a^2 + 7 \cdot M_b^2 + 4 \cdot M_c^2}}\right) = 1.131$$

$$M_u := \frac{\omega_2 \cdot \pi}{L} \cdot \sqrt{E_s \cdot I_y \cdot G_s \cdot J + \left(\frac{\pi \cdot E_s}{L}\right)^2 \cdot I_y \cdot C_w} = 133.59 \cdot kN \cdot m$$

critical elastic moment

$$M_{rx.nolat} := \begin{array}{|l} \text{if } Section_{class} = 1 \vee 2 \\ \quad \begin{array}{|l} 1.15 \cdot \phi_s \cdot Z_x \cdot F_y \cdot \left(1 - \frac{0.28 \cdot Z_x \cdot F_y}{M_u}\right) \text{ if } M_u > 0.67 \cdot Z_x \cdot F_y \\ \phi_s \cdot M_u \text{ otherwise} \end{array} \\ \text{if } Section_{class} = 3 \\ \quad \begin{array}{|l} 1.15 \cdot \phi_s \cdot S_x \cdot F_y \cdot \left(1 - \frac{0.28 \cdot S_x \cdot F_y}{M_u}\right) \text{ if } M_u > 0.67 \cdot S_x \cdot F_y \\ \phi_s \cdot M_u \text{ otherwise} \end{array} \end{array} = 108.95 \cdot kN \cdot m$$

**PURLIN AND GIRT SIZING CALCULATION**

$$UC_{nolateral.support} := \frac{M_{fx.nolat}}{M_{rx.nolat}} + \frac{M_{fy.nolat}}{M_{ry}} = 0.824$$ less than 1 OK

Check Deflection

$$\Delta_y := \frac{5 \cdot \max\left(L_{y.pressure.SLS}, L_{y.suction.SLS}\right) \cdot (L)^4}{384 \cdot E_s \cdot I_x} = 8.84 \cdot mm$$

$$\Delta_x := \frac{5 \cdot L_{x.SLS} \cdot L^4}{384 \cdot E_s \cdot I_y} = 16.32 \cdot mm$$

$$\Delta_{allowable} := \frac{L}{180} = 38.89 \cdot mm$$ greater than x,y deflection OK

**PURLIN AND GIRT SIZING CALCULATION**

9.4 Roof/Wall Panel Check

For Roof Panel

Design Loads (SLS): Wind Load - $WL_{r.suction} := 2.5kPa$ $WL_{r.suction} = 52.21 \cdot psf$

$WL_{r.pressure} := 0.35kPa$ $WL_{r.pressure} = 7.31 \cdot psf$

Snow Load - $q_{snow} := 1.24kPa$ $q_{snow} = 25.9 \cdot psf$

Purlin Spacing: $s_{purlin} := 1.5m$ $s_{purlin} = 4.92 \cdot ft$

The table presented below show the panel weight, allowable load and span for panel materials aproved by the project as per PDH-00-AR-SPC-0001-10 (Insulated Composite Metal Panels)

$roof_{panel.bldg} :=$

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | "Kingspan" | "AWIP" | "Metlspan" |
| 1 | "panel wt (psf)" | 3.05 | 2.65 | 0 |
| 2 | "AL for Wind (psf)"" | 69.46 | 81 | 51.8 |
| 3 | "AL for Snow (psf)" | 67.11 | 87 | 93 |
| 4 | "span (ft)" | 5 | 5 | 5 |

Note: 1. Roof panel is 4" thick.
2. AL is Allowable Load.
3. Allowable load for Wind is governed by connection capacity.
4. Allowable load for Snow is governed by deflection or bending/shear of panel.
5. Connection capacity is for 2 fasteners.
6. Panel weight information for Meltspan is not available. Assumed similar to Kingspan for DL purposes.
7. Effect of panel weight is assumed not to be included in the tabulated allowable loads.
8. Refer to Appendix for Manufacturers' product information.

$$panel_{wt} := \max\left(roof_{panel.bldg_{1,1}}, roof_{panel.bldg_{1,2}}, roof_{panel.bldg_{1,3}}\right) \cdot psf = 3.05 \cdot psf$$

Maximum allowable load for wind is

$$WL_{r.allowable} := \min\left(roof_{panel.bldg_{2,1}}, roof_{panel.bldg_{2,2}}, roof_{panel.bldg_{2,3}}\right) \cdot psf = 51.8 \cdot psf$$

$$checkpanel_{r.for.wind} := \begin{cases} \text{"Panel is OK"} & \text{if } WL_{r.allowable} > WL_{r.suction} \\ \text{"Adjust purlin spacing"} & \text{otherwise} \end{cases} = \text{"Adjust purlin spacing"}$$

say OK - can use more than 2 fasteners to increase connection capacity

Maximum allowable load for snow is

$$SL_{allowable} := \min\left(roof_{panel.bldg_{3,1}}, roof_{panel.bldg_{3,2}}, roof_{panel.bldg_{3,3}}\right) \cdot psf = 67.11 \cdot psf$$

$$checkpanel_{for.snow} := \begin{cases} \text{"Panel is OK"} & \text{if } SL_{allowable} > q_{snow} + panel_{wt} \\ \text{"Adjust purlin spacing"} & \text{otherwise} \end{cases} = \text{"Panel is OK"}$$

## PURLIN AND GIRT SIZING CALCULATION

For Wall Panel

Design Loads (SLS): Wind Load - $WL_{w.suction} := 1.42kPa$ $WL_{w.suction} = 29.66 \cdot psf$

$WL_{w.pressure} := 1.24kPa$ $WL_{w.pressure} = 25.9 \cdot psf$

Girt Spacing: $s_{girt} := 1.8m$ $s_{girt} = 5.91 \cdot ft$

The table presented below show the panel weight, allowable load and span for non-fire rated panel materials aproved by the project as per PDH-00-AR-SPC-0001-10 (Insulated Composite Metal Panels)

$wall_{panel} :=$

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | "Kingspan" | "AWIP" | "Metlspan" |
| 1 | "panel wt (psf)" | 2.86 | 2.41 | "N/A" |
| 2 | "AL for Wind (psf)" | 30 | 27 | 28.8 |
| 3 | "span (ft)" | 6.83 | 6 | 6 |

Note: 1. Wall panel is 3" thick.
2. AL is Allowable Load.
3. Allowable load for Wind is governed by connection capacity.
4. Connection capacity is for 2 fasteners.
5. Refer to Appendix for Manufacturers' product information.

Maximum allowable load for wind is

$$WL_{w.allowable} := \min\left(wall_{panel_{2,1}}, wall_{panel_{2,2}}, wall_{panel_{2,3}}\right) \cdot psf = 27 \cdot psf$$

$$checkpanel_{w.for.wind} := \begin{vmatrix} \text{"Panel is OK"} \quad \text{if } WL_{w.allowable} > WL_{w.suction} \\ \text{"Adjust purlin spacing"} \quad \text{otherwise} \end{vmatrix} = \text{"Adjust purlin spacing"}$$

*Connection capacity can be increased by using more than 2 fasteners.

The table presented below show the panel weight, allowable load and span for fire rated panel materials aproved by the project as per PDH-00-AR-SPC-0001-10 (Insulated Composite Metal Panels)

$wall_{panel.FR} :=$

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | "Kingspan" | "AWIP" | "Meltspan" |
| 1 | "panel wt (psf)" | 5.2 | 5.5 | "N/A" |
| 2 | "allowable load (psf)" | 40.6 | 40 | 49.6 |
| 3 | "span (ft)" | 6 | 6 | 6 |

Note: 1. Wall panel is 4" thick.4.5" for AWIP fiRe.
2. AL is Allowable Load.
3. Allowable load for Wind is governed by connection capacity.
4. Connection capacity is for 3 fasteners.
5. Refer to Appendix for Manufacturers' product information.
6. Allowable load for Kingspan fiRe Panel is based on 4" standard as per manufacturer's guidance.

## PURLIN AND GIRT SIZING CALCULATION

$$WL_{w.allowable.FR} := \min\left(wall_{panel.FR_{2,1}}, wall_{panel.FR_{2,2}}, wall_{panel.FR_{2,3}}\right)\cdot psf = 40\cdot psf$$

$$checkpanel_{w.for.wind.FR} := \begin{array}{|l} \text{"Panel is OK"} \quad \text{if } WL_{w.allowable.FR} > WL_{w.suction} \\ \text{"Adjust purlin spacing"} \quad \text{otherwise} \end{array} = \text{"Panel is OK"}$$

**PURLIN AND GIRT SIZING CALCULATION**

## 10..Purlind and Girt Design for Stair Tower Module

10.1 Purlin Design: Stair Tower - Load and height is based on 02HP Stair Tower (governing module)

Purlin Section - W200x36 Purlin Spacing - $s := 0.8m$ Purlin Length - $L := 6m$

Note: Purlin spacing is based on critical purlin. Refer to page 12

Section Properties:

$wt := 0.352\frac{kN}{m}$ section dead load

$b := 165mm$ flange width

$d := 201mm$ depth

$t_w := 6.2mm$ web thickness

$t_f := 10.2mm$ flange thickness

$I_x := 34.4\cdot 10^6 mm^4$ $I_y := 7.64\cdot 10^6 mm^4$ $J := 145\cdot 10^3\cdot mm^4$

$S_x := 342\cdot 10^3 mm^3$ $S_y := 92.6\cdot 10^3 mm^3$ $C_w := 69.6\cdot 10^9\cdot mm^6$

$Z_x := 379\cdot 10^3 mm^3$ $Z_y := 141\cdot 10^3 mm^3$

$F_y := 350MPa$ $E_s := 200GPa$ $G_s := 77GPa$ $\phi_s := 0.9$

Roof Details: $\theta := 9.462deg$

4" Kingzip Standing Seam Panels 24 Ga (governing panel) - as per PDH-00-AR-SPC-0001-10

$wt_{panel} := 3.05psf = 0.146\,kPa$

Loads: $q_{live} := 1kPa$ Live load

$q_e := 1kPa$ Equipment/Electrical Load

$q_{snow} := 7.08kPa$ Snow Load (ULS) - refer to page 10

Wind Load Calculation for Purlins:

$I_{ULS} := 1.15$ $I_{SLS} := 0.75$ Importance Factor

$q := 0.43kPa$ reference velocity pressure

$El_{roof} := 119.2m$ roof elevation

$El_{grade} := 100m$ grade elevation

$$C_e := \left(\frac{El_{roof} - El_{grade}}{10m}\right)^{0.2} = 1.14$$ exposure factor

$C_g := 2.5$ external gust effect factor for small elements incl. cladding

$C_p := 2.3$ external suction coeefficient for roofing and cladding

## PURLIN AND GIRT SIZING CALCULATION

$C_{gi} := 2$ internal gust effect factor

$C_{pi.s} := 0.45$ internal suction coefficient

$C_{pi.p} := 0.3$ internal presure coefficient

$WL_{external.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_p = 3.24 \cdot kPa$ for External Suction

$WL_{internal.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.51 \cdot kPa$ for Internal Suction

$WL_{internal.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.34 \cdot kPa$ for Internal Pressure

$WL_{external.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_p = 2.11 \cdot kPa$ for External Suction

$WL_{internal.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.33 \cdot kPa$ for Internal Suction

$WL_{internal.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.22 \cdot kPa$ for Internal Pressure

Total Wind Load:

Downward Force - $WL_{ULS.down} := WL_{internal.s.ULS} = 0.51 \cdot kPa$ $WL_{ULS.down} = 10.59 \cdot psf$

$WL_{SLS.down} := WL_{internal.s.SLS} = 0.33 \cdot kPa$ $WL_{SLS.down} = 6.907 \cdot psf$

Upward Force - $WL_{ULS.up} := WL_{external.ULS} + WL_{internal.p.ULS} = 3.58 \cdot kPa$ $WL_{ULS.up} = 74.721 \cdot psf$

$WL_{SLS.up} := WL_{external.SLS} + WL_{internal.p.SLS} = 2.33 \cdot kPa$ $WL_{SLS.up} = 48.731 \cdot psf$

Factored Linear load on purlins

Load Combination 1: 1.25DL + 1.5SL +0.5LL

$$L_{y1.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta) + 0.5 \cdot q_{live} \cdot s\right] \cdot \cos(\theta) = 10.23 \cdot \frac{kN}{m}$$

$$L_{x1.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta) + 0.5 \cdot q_{live} \cdot s\right] \cdot \sin(\theta) = 1.7 \cdot \frac{kN}{m}$$

Load Combination 2: 1.25DL + 1.5SL + 0.4WLdown

$$L_{y2.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \cos(\theta) + 0.4 \cdot WL_{ULS.down} \cdot s = 9.99 \cdot \frac{kN}{m}$$

$$L_{x2.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 1.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \sin(\theta) = 1.64 \cdot \frac{kN}{m}$$

Load Combination 3: 1.25DL + 1.4WLdown + 0.5SL

$$L_{y3.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 0.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \cos(\theta) + 1.4 \cdot WL_{ULS.down} \cdot s = 4.888 \cdot \frac{kN}{m}$$

$$L_{x3.ULS} := \left[1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + 0.5 \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \sin(\theta) = 0.72 \cdot \frac{kN}{m}$$

Load Combination 4: 1.25DL - 1.4WLup - does not include equipment weight which is conservative

$$L_{y4.ULS} := 1.25 \cdot \left(wt + wt_{panel} \cdot s\right) \cdot \cos(\theta) - 1.4 \cdot WL_{ULS.up} \cdot s = -3.429 \cdot \frac{kN}{m}$$

$$L_{x4.ULS} := 1.25 \cdot \left(wt + wt_{panel} \cdot s\right) \sin(\theta) = 0.096 \cdot \frac{kN}{m}$$

## PURLIN AND GIRT SIZING CALCULATION

Governing Factored Load: $y_{ULS} := \left(L_{y1.ULS}\ \ L_{y2.ULS}\ \ L_{y3.ULS}\ \ L_{y4.ULS}\right)$

$x_{ULS} := \left(L_{x1.ULS}\ \ L_{x2.ULS}\ \ L_{x3.ULS}\ \ L_{x4.ULS}\right)$

$L_{y.ULS.down} := \max\left(y_{ULS}\right) = 10.23 \cdot \frac{kN}{m}$ $L_{y.ULS.up} := \min\left(y_{ULS}\right) = -3.43 \cdot \frac{kN}{m}$ $L_{x.ULS} := \max\left(x_{ULS}\right) = 1.7 \cdot \frac{kN}{m}$

Service Linear Load on purlins

Load Combination 1: DL + SL +LL

$$L_{y1.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta) + q_{live} \cdot s\right] \cdot \cos(\theta) = 6.35 \cdot \frac{kN}{m}$$

$$L_{x1.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta) + q_{live} \cdot s\right] \cdot \sin(\theta) = 1.06 \cdot \frac{kN}{m}$$

Load Combination 2: DL + SL + WLdown

$$L_{y2.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \cos(\theta) + 0.4 \cdot WL_{SLS.down} \cdot s = 5.67 \cdot \frac{kN}{m}$$

$$L_{x2.SLS} := \left[\left(wt + wt_{panel} \cdot s + q_e \cdot s\right) + \frac{0.9}{1.15} \cdot q_{snow} \cdot s \cdot \cos(\theta)\right] \cdot \sin(\theta) = 0.93 \cdot \frac{kN}{m}$$

Load Combination 3: DL - WLup - does not include equipment weight which is conservative

$$L_{y3.SLS} := \left(wt + wt_{panel} \cdot s\right) \cdot \cos(\theta) - WL_{SLS.up} \cdot s = -1.404 \cdot \frac{kN}{m}$$

$$L_{x3.SLS} := \left(wt + wt_{panel} \cdot s\right) \cdot \sin(\theta) = 0.077 \cdot \frac{kN}{m}$$

Governing Service Load: $y_{SLS} := \left(L_{y1.SLS}\ \ L_{y2.SLS}\ \ L_{y3.SLS}\right)$

$x_{SLS} := \left(L_{x1.SLS}\ \ L_{x2.SLS}\ \ L_{x3.SLS}\right)$

$$L_{y.SLS.down} := \max\left(y_{SLS}\right) = 6.35 \cdot \frac{kN}{m}$$

$$L_{y.SLS.up} := \min\left(y_{SLS}\right) = -1.4 \cdot \frac{kN}{m}$$

$$L_{x.SLS} := \max\left(x_{SLS}\right) = 1.06 \cdot \frac{kN}{m}$$

## PURLIN AND GIRT SIZING CALCULATION

### Purlin Design

Section Class

$$Flange_{class} := \begin{cases} 1 & \text{if } \dfrac{b}{2\cdot t_f} \le \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{200}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 2$$

$$Web_{class} := \begin{cases} 1 & \text{if } \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1900}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{cases} = 1$$

$$Section_{class} := \max\left(Flange_{class}, Web_{class}\right) = 2$$

Check Bending Capacity

*For laterally supported members:* This is applicable to purlins when force direction is downward. During this case the top flange is in compression and the roof panel provides lateral support to the compression flange.
For moment capacity in weak direction: Since load is applied on top flange, only half the section capacity is considered for design.

$$M_{fx.down} := \left|\frac{L_{y.ULS.down}\cdot L^2}{8}\right| = 46.01\cdot kN\cdot m$$

$$M_{fy.down} := \left|\frac{L_{x.ULS}\cdot L^2}{8}\right| = 7.67\cdot kN\cdot m$$

$$M_{rx.lat} := \begin{cases} \phi_s\cdot Z_x\cdot F_y & \text{if } Section_{class} = 1 \vee 2 \\ \phi_s\cdot S_x\cdot F_y & \text{if } Section_{class} = 3 \\ \text{"use Class 1,2 or 3"} & \text{otherwise} \end{cases} = 119.385\cdot kN\cdot m$$

$$M_{ry} := \frac{1}{2}\cdot\left(\phi_s\cdot S_y\cdot F_y\right) = 14.58\cdot kN\cdot m$$

$$UC_{lateral} := \frac{M_{fx.down}}{M_{rx.lat}} + \frac{M_{fy.down}}{M_{ry}} = 0.911$$ less than 1 OK

## PURLIN AND GIRT SIZING CALCULATION

*For laterally unsupported members:* This is applicable to purlins when force direction is upward. During this case the bottom flange is in compression and no lateral restraint is provided to prevent lateral buckling.
For moment capacity in weak direction: Since load is applied on top flange, only half the section capacity is considered for design.

$$M_{fx.up} := \left| \frac{L_{y.ULS.up} \cdot L^2}{8} \right| = 15.43 \cdot kN \cdot m$$

$$M_{fy.up} := \left| \frac{\text{lookup}(L_{y.ULS.up}, y_{ULS}, x_{ULS}) \cdot L^2}{8} \right| = 0.43 \cdot kN \cdot m$$

$$M_{max} := M_{fx.up} = 15.43 \cdot kN \cdot m$$

$$M_a := \left| L_{y.ULS.up} \cdot \frac{3L^2}{32} \right| = 11.573 \cdot kN \cdot m$$

factored bending moment at 1/4 point of unbraced segment

$$M_b := M_{fx.up} = 15.43 \cdot kN \cdot m$$

factored bending moment at midpoint of unbraced segment

$$M_c := \left| L_{y.ULS.up} \cdot \frac{3L^2}{32} \right| = 11.573 \cdot kN \cdot m$$

factored bending moment at 3/4 point of unbraced segment

$$\omega_2 := \min\left(2.5, \frac{4 \cdot M_{max}}{\sqrt{M_{max}^2 + 4 \cdot M_a^2 + 7 \cdot M_b^2 + 4 \cdot M_c^2}}\right) = 1.131$$

$$M_u := \frac{\omega_2 \cdot \pi}{L} \cdot \sqrt{E_s \cdot I_y \cdot G_s \cdot J + \left(\frac{\pi \cdot E_s}{L}\right)^2 \cdot I_y \cdot C_w} = 89.63 \cdot kN \cdot m$$

critical elastic moment

$$M_{rx.nolat} := \begin{cases} \text{if } Section_{class} = 1 \vee 2 \\ \quad \begin{cases} 1.15 \cdot \phi_s \cdot Z_x \cdot F_y \cdot \left(1 - \frac{0.28 \cdot Z_x \cdot F_y}{M_u}\right) & \text{if } M_u > 0.67 \cdot Z_x \cdot F_y \\ \phi_s \cdot M_u & \text{otherwise} \end{cases} \\ \text{if } Section_{class} = 3 \\ \quad \begin{cases} 1.15 \cdot \phi_s \cdot S_x \cdot F_y \cdot \left(1 - \frac{0.28 \cdot S_x \cdot F_y}{M_u}\right) & \text{if } M_u > 0.67 \cdot S_x \cdot F_y \\ \phi_s \cdot M_u & \text{otherwise} \end{cases} \end{cases} = 80.4 \cdot kN \cdot m$$

## PURLIN AND GIRT SIZING CALCULATION

$$UC_{nolateral.support} := \frac{M_{fx.up}}{M_{rx.nolat}} + \frac{M_{fy.up}}{M_{ry}} = 0.222$$ less than 1 OK

Check Deflection

$$\Delta_y := \frac{5 \cdot \max\left(L_{y.SLS.down}, L_{y.SLS.up}\right) \cdot (L)^4}{384 \cdot E_s \cdot I_x} = 15.58 \cdot mm$$

$$\Delta_x := \frac{5 \cdot L_{x.SLS} \cdot (L - 0.8m)^4}{384 \cdot E_s \cdot I_y} = 6.6 \cdot mm$$

$$\Delta_{allowable} := \frac{L}{240} = 25 \cdot mm$$ greater than x,y deflection OK

## PURLIN AND GIRT SIZING CALCULATION

10.2 Girt Design: Stair Tower - Load and height is based on 02HP Stair Tower (governing module)

Girt Section - W200x36 Girt Spacing - $s := 1.8m$ Girt Length - $L := 6m$

Section Properties:

$wt := 0.352 \frac{kN}{m}$ section dead load

$b := 165mm$ flange width

$d := 201mm$ depth

$t_w := 6.2mm$ web thickness

$t_f := 10.2mm$ flange thickness

$I_x := 34.4 \cdot 10^6 mm^4$ $I_y := 7.64 \cdot 10^6 mm^4$ $J := 145 \cdot 10^3 \cdot mm^4$

$S_x := 342 \cdot 10^3 mm^3$ $S_y := 92.6 \cdot 10^3 mm^3$ $C_w := 69.6 \cdot 10^9 \cdot mm^6$

$Z_x := 379 \cdot 10^3 mm^3$ $Z_y := 141 \cdot 10^3 mm^3$

$F_y := 350MPa$ $E_s := 200GPa$ $G_s := 77GPa$ $\phi_s := 0.9$

6" Fire Rated Panel Kingspan Shadowline MF 24 Ga (governing panel) - as per PDH-00-AR-SPC-0001-10

$wt_{panel} := 6.7psf = 0.321 \cdot kPa$

Loads: Wind Load Calculation for Girts:

$I_{ULS} := 1.15$ $I_{SLS} := 0.75$ Importance Factor

$q := 0.43kPa$ reference velocity pressure

$El_{roof} := 119.2m$ roof elevation

$El_{grade} := 100m$ grade elevation

$C_e := \left(\frac{El_{roof} - El_{grade}}{10m}\right)^{0.2} = 1.14$ exposure factor

$C_g := 2.5$ external gust effect factor for small elements incl. cladding

$C_{p.s} := 1.2$ external suction coefficient for cladding

$C_{p.p} := 0.9$ external pressure coefficient for cladding

$C_{gi} := 2$ internal gust effect factor

$C_{pi.s} := 0.45$ internal suction coefficient

**PURLIN AND GIRT SIZING CALCULATION**

$C_{pi.p} := 0.3$ internal presure coefficient

ULS Load:

$WL_{external.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.s} = 1.69 \cdot kPa$ for External Suction

$WL_{external.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.p} = 1.27 \cdot kPa$ for External Pressure

$WL_{internal.s.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.51 \cdot kPa$ for Internal Suction

$WL_{internal.p.ULS} := I_{ULS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.34 \cdot kPa$ for Internal Pressure

SLS Load:

$WL_{external.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.s} = 1.102 \cdot kPa$ for External Suction

$WL_{external.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_g \cdot C_{p.p} = 0.83 \cdot kPa$ for External Pressure

$WL_{internal.s.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.s} = 0.331 \cdot kPa$ for Internal Suction

$WL_{internal.p.SLS} := I_{SLS} \cdot q \cdot C_e \cdot C_{gi} \cdot C_{pi.p} = 0.22 \cdot kPa$ for Internal Pressure

Total Wind Load:

Total Wind Load:

Wall Suction (Negative WL) - $WL_{suction.ULS} := WL_{external.s.ULS} + WL_{internal.p.ULS} = 2.03 \cdot kPa$

$WL_{suction.SLS} := WL_{external.s.SLS} + WL_{internal.p.SLS} = 1.32 \cdot kPa$

Wall Pressure (Positive WL) - $WL_{pressure.ULS} := WL_{external.p.ULS} + WL_{internal.s.ULS} = 1.77 \cdot kPa$

$WL_{pressure.SLS} := WL_{external.p.SLS} + WL_{internal.s.SLS} = 1.16 \cdot kPa$

$q_e := 0.5kPa$ additional load due to equipment/tray/pipe supported on girts

Factored Linear load on girts

<u>Load Combination: 1.25DL + 1.4WL</u>

$$L_{x.ULS} := 1.25 \cdot \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) = 2.29 \cdot \frac{kN}{m}$$

$$L_{y.suction.ULS} := 1.4 \cdot WL_{suction.ULS} \cdot s = 5.11 \cdot \frac{kN}{m}$$

$$L_{y.pressure.ULS} := 1.4 \cdot WL_{pressure.ULS} \cdot s = 4.47 \cdot \frac{kN}{m}$$

Service Linear Load on girts

<u>Load Combination: DL + WL</u>

$$L_{x.SLS} := \left(wt + wt_{panel} \cdot s + q_e \cdot s\right) = 1.83 \cdot \frac{kN}{m}$$

$$L_{y.suction.SLS} := WL_{suction.SLS} \cdot s = 2.38 \cdot \frac{kN}{m}$$

$$L_{y.pressure.SLS} := WL_{pressure.SLS} \cdot s = 2.08 \cdot \frac{kN}{m}$$

**Girt Design**

Section Class

$$\text{Flange}_{class} := \left| \begin{array}{ll} 1 & \text{if } \dfrac{b}{2\cdot t_f} \le \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{145}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{170}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{b}{2\cdot t_f} \le \dfrac{200}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{array} \right. = 2$$

$$\text{Web}_{class} := \left| \begin{array}{ll} 1 & \text{if } \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} \\ 2 & \text{if } \dfrac{1100}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} \\ 3 & \text{if } \dfrac{1700}{\sqrt{\dfrac{F_y}{MPa}}} < \dfrac{d - 2\cdot t_f}{t_w} \le \dfrac{1900}{\sqrt{\dfrac{F_y}{MPa}}} \\ 4 & \text{otherwise} \end{array} \right. = 1$$

$$\text{Section}_{class} := \max\left(\text{Flange}_{class}, \text{Web}_{class}\right) = 2$$

Check Bending Capacity

*For laterally supported members:* This is applicable to girts when flange with wall connection is in compression. During this case the flange is in compression and the wall panel provides lateral support to the flange.
For moment capacity in weak direction: Only half the section capacity is considered for design.

$$M_{fx.lat} := \left| \dfrac{L_{y.pressure.ULS}\cdot L^2}{8} \right| = 20.13\cdot kN\cdot m$$

$$M_{fy.lat} := \left| \dfrac{L_{x.ULS}\cdot L^2}{8} \right| = 10.29\cdot kN\cdot m$$

$$M_{rx.lat} := \left| \begin{array}{ll} \phi_s\cdot Z_x\cdot F_y & \text{if Section}_{class} = 1 \vee 2 \\ \phi_s\cdot S_x\cdot F_y & \text{if Section}_{class} = 3 \\ \text{"use Class 1,2 or 3"} & \text{otherwise} \end{array} \right. = 119.385\cdot kN\cdot m$$

$$M_{ry} := \frac{1}{2}\cdot\left(\phi_s\cdot S_y\cdot F_y\right) = 14.58\cdot kN\cdot m$$

$$UC_{lateral} := \frac{M_{fx.lat}}{M_{rx.lat}} + \frac{M_{fy.lat}}{M_{ry}} = 0.874$$ less than 1 OK

*For laterally unsupported members:* This is applicable to girts when flange with wall connection is in tension During this case the flange is in tension and no lateral restraint is provided to prevent lateral buckling.
For moment capacity in weak direction: Only half the section capacity is considered for design.

$$M_{fx.nolat} := \left| \frac{L_{y.suction.ULS} \cdot L^2}{8} \right| = 23 \cdot kN \cdot m$$

$$M_{fy.nolat} := \left| \frac{L_{x.ULS} \cdot L^2}{8} \right| = 10.29 \cdot kN \cdot m$$

$$M_{max} := M_{fx.nolat} = 23 \cdot kN \cdot m$$

$$M_a := \left| L_{y.suction.ULS} \cdot \frac{3L^2}{32} \right| = 17.3 \cdot kN \cdot m$$

factored bending moment at 1/4 point of unbraced segment

$$M_b := M_{fx.nolat} = 23 \cdot kN \cdot m$$

factored bending moment at midpoint of unbraced segment

$$M_c := \left| L_{y.suction.ULS} \cdot \frac{3L^2}{32} \right| = 17.25 \cdot kN \cdot m$$

factored bending moment at 3/4 point of unbraced segment

$$\omega_2 := \min\left( 2.5, \frac{4 \cdot M_{max}}{\sqrt{{M_{max}}^2 + 4 \cdot {M_a}^2 + 7 \cdot {M_b}^2 + 4 \cdot {M_c}^2}} \right) = 1.131$$

$$M_u := \frac{\omega_2 \cdot \pi}{L} \cdot \sqrt{E_s \cdot I_y \cdot G_s \cdot J + \left( \frac{\pi \cdot E_s}{L} \right)^2 \cdot I_y \cdot C_w} = 89.63 \cdot kN \cdot m$$

critical elastic moment

$$M_{rx.nolat} := \begin{array}{l} \text{if } Section_{class} = 1 \vee 2 \\ \quad \begin{array}{|l} 1.15 \cdot \phi_s \cdot Z_x \cdot F_y \cdot \left( 1 - \frac{0.28 \cdot Z_x \cdot F_y}{M_u} \right) \text{ if } M_u > 0.67 \cdot Z_x \cdot F_y \\ \phi_s \cdot M_u \text{ otherwise} \end{array} \\ \text{if } Section_{class} = 3 \\ \quad \begin{array}{|l} 1.15 \cdot \phi_s \cdot S_x \cdot F_y \cdot \left( 1 - \frac{0.28 \cdot S_x \cdot F_y}{M_u} \right) \text{ if } M_u > 0.67 \cdot S_x \cdot F_y \\ \phi_s \cdot M_u \text{ otherwise} \end{array} \end{array} = 80.4 \cdot kN \cdot m$$

$$UC_{nolateral.support} := \frac{M_{fx.nolat}}{M_{rx.nolat}} + \frac{M_{fy.nolat}}{M_{ry}} = 0.992$$ less than 1 OK

Check Deflection

$$\Delta_y := \frac{5 \cdot \max\left(L_{y.pressure.SLS}, L_{y.suction.SLS}\right) \cdot (L)^4}{384 \cdot E_s \cdot I_x} = 5.84 \cdot mm$$

$$\Delta_x := \frac{5 \cdot L_{x.SLS} \cdot (L - 0.8m)^4}{384 \cdot E_s \cdot I_y} = 11.4 \cdot mm$$

$$\Delta_{allowable} := \frac{L}{180} = 33.33 \cdot mm$$ greater than x,y deflection OK

10.3 Roof/Wall Panel Check

For Roof Panel

Design Loads (SLS): Wind Load - $WL_{r.suction} := 2.33kPa$  $WL_{r.suction} = 48.66 \cdot psf$

$WL_{r.pressure} := 0.33kPa$  $WL_{r.pressure} = 6.89 \cdot psf$

Snow Load - $q_{snow} := 5.54kPa$  $q_{snow} = 115.71 \cdot psf$

Purlin Spacing: $s_{purlin} := 0.7m$  $s_{purlin} = 2.3 \cdot ft$

The table presented below show the panel weight, allowable load and span for panel materials aproved by the project as per PDH-00-AR-SPC-0001-10 (Insulated Composite Metal Panels)

$roof_{panel.bldg} :=$

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | "Kingspan" | "AWIP" | "Metlspan" |
| 1 | "panel wt (psf)" | 3.05 | 2.65 | 0 |
| 2 | "AL for Wind (psf)"" | 131.02 | 160 | 84 |
| 3 | "AL for Snow (psf)" | 138.26 | 185 | 196.2 |
| 4 | "span (ft)" | 2.5 | 2.5 | 2.5 |

Note: 1. Roof panel is 4" thick.
2. AL is Allowable Load.
3. Allowable load for Wind is governed by connection capacity.
4. Allowable load for Snow is governed by deflection or bending/shear of panel.
5. Connection capacity is for 2 fasteners.
6. Panel weight information for Meltspan is not available.
7. Effect of panel weight is assumed not to be included in the tabulated allowable loads.
8. Refer to Appendix for Manufacturers' product information.

$$panel_{wt} := \max\left(roof_{panel.bldg_{1,1}}, roof_{panel.bldg_{1,2}}, roof_{panel.bldg_{1,3}}\right) \cdot psf = 3.05 \cdot psf$$

Maximum allowable load for wind is

$$WL_{r.allowable} := \min\left(roof_{panel.bldg_{2,1}}, roof_{panel.bldg_{2,2}}, roof_{panel.bldg_{2,3}}\right) \cdot psf = 84 \cdot psf$$

$$checkpanel_{r.for.wind} := \begin{array}{|l} \text{"Panel is OK"} \quad \text{if } WL_{r.allowable} > WL_{r.suction} \\ \text{"Adjust purlin spacing"} \quad \text{otherwise} \end{array} = \text{"Panel is OK"}$$

Maximum allowable load for snow is

$$SL_{allowable} := \min\left(roof_{panel.bldg_{3,1}}, roof_{panel.bldg_{3,2}}, roof_{panel.bldg_{3,3}}\right) \cdot psf = 138.26 \cdot psf$$

$$checkpanel_{for.snow} := \begin{array}{|l} \text{"Panel is OK"} \quad \text{if } SL_{allowable} > q_{snow} + panel_{wt} \\ \text{"Adjust purlin spacing"} \quad \text{otherwise} \end{array} = \text{"Panel is OK"}$$

For Wall Panel

Design Loads (SLS): Wind Load - $WL_{w.suction} := 1.32kPa$ $WL_{w.suction} = 27.57 \cdot psf$

$WL_{w.pressure} := 1.16kPa$ $WL_{w.pressure} = 24.23 \cdot psf$

Girt Spacing: $s_{girt} := 1.8m$ $s_{girt} = 5.91 \cdot ft$

The table presented below show the panel weight, allowable load and span for fire rated panel materials aproved by the project as per PDH-00-AR-SPC-0001-10 (Insulated Composite Metal Panels)

$wall_{panel.FR} :=$

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | "Kingspan" | "AWIP" | "Metlspan" |
| 1 | "panel wt (psf)" | 6.7 | 5.5 | "N/A" |
| 2 | "allowable load (psf)" | 41.6 | 40 | 50.9 |
| 3 | "span (ft)" | 6 | 6 | 6 |

Note: 1. Wall panel is 6" thick. 4.5" for AWIP fiRe
2. AL is Allowable Load.
3. Allowable load for Wind is governed by connection capacity.
4. Connection capacity is for 3 fasteners.
5. Refer to Appendix for Manufacturers' product information.

$$WL_{w.allowable.FR} := \min\left(wall_{panel.FR_{2,1}}, wall_{panel.FR_{2,2}}, wall_{panel.FR_{2,3}}\right) \cdot psf = 40 \cdot psf$$

$$checkpanel_{w.for.wind.FR} := \begin{array}{|l} \text{"Panel is OK" if } WL_{w.allowable.FR} > WL_{w.suction} \\ \text{"Adjust purlin spacing" otherwise} \end{array} = \text{"Panel is OK"}$$

Roof Panel loading check for Stair Tower with 1m spacing (located at low eave of Stair Tower):

Design Loads (SLS): Wind Load - $WL_{r.suction} := 2.33kPa$ $WL_{r.suction} = 48.66 \cdot psf$

$WL_{r.pressure} := 0.33kPa$ $WL_{r.pressure} = 6.89 \cdot psf$

Snow Load - $q_{snow} := 5kPa$ $q_{snow} = 104.43 \cdot psf$ considering average snow load

Purlin Spacing: $s_{purlin} := 1m$ $s_{purlin} = 3.28 \cdot ft$

The table presented below show the panel weight, allowable load and span for panel materials aproved by the project as per PDH-00-AR-SPC-0001-10 (Insulated Composite Metal Panels)

$roof_{panel.bldg} :=$

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | "Kingspan" | "AWIP" | "Metlspan" |
| 1 | "panel wt (psf)" | 3.05 | 2.65 | 0 |
| 2 | "AL for Wind (psf)"" | 95.83 | 116 | 59.9 |
| 3 | "AL for Snow (psf)" | 97.75 | 130 | 118.6 |
| 4 | "span (ft)" | 3.5 | 3.5 | 3.5 |

Note: 1. Roof panel is 4" thick.
2. AL is Allowable Load.
3. Allowable load for Wind is governed by connection capacity.
4. Allowable load for Snow is governed by deflection or bending/shear of panel.
5. Connection capacity is for 2 fasteners.
6. Panel weight information for Meltspan is not available.
7. Effect of panel weight is assumed not to be included in the tabulated allowable loads.
8. Refer to Appendix for Manufacturers' product information.

$$panel_{wt} := \max\left(roof_{panel.bldg_{1,1}}, roof_{panel.bldg_{1,2}}, roof_{panel.bldg_{1,3}}\right) \cdot psf = 3.05 \cdot psf$$

Maximum allowable load for wind is

$$WL_{r.allowable} := \min\left(roof_{panel.bldg_{2,1}}, roof_{panel.bldg_{2,2}}, roof_{panel.bldg_{2,3}}\right) \cdot psf = 59.9 \cdot psf$$

$$checkpanel_{r.for.wind} := \begin{cases} \text{"Panel is OK"} & \text{if } WL_{r.allowable} > WL_{r.suction} \\ \text{"Adjust purlin spacing"} & \text{otherwise} \end{cases} = \text{"Panel is OK"}$$

Maximum allowable load for snow is

$$SL_{allowable} := \min\left(roof_{panel.bldg_{3,1}}, roof_{panel.bldg_{3,2}}, roof_{panel.bldg_{3,3}}\right) \cdot psf = 97.75 \cdot psf$$

$$checkpanel_{for.snow} := \begin{cases} \text{"Panel is OK"} & \text{if } SL_{allowable} > q_{snow} + panel_{wt} \\ \text{"Adjust purlin spacing"} & \text{otherwise} \end{cases} = \text{"Adjust purlin spacing"}$$

Governing SL allowable is for Kingspan at 3.5ft spacing. By interpolation allowable SL at 1m spacing is:

$$x := \begin{pmatrix} 3\text{ft} \\ 3.5\text{ft} \\ 4\text{ft} \end{pmatrix} \quad y := \begin{pmatrix} 114.67\text{psf} \\ 97.75\text{psf} \\ 85.02\text{psf} \end{pmatrix} \quad \text{load} := \text{linterp}\left(x, y, s_{purlin}\right) = 105.17\,\text{psf}$$

Allowable SL for 1m spacing

Therefore, panel is ok for 1m spacing.

**PURLIN AND GIRT SIZING CALCULATION**

## 11. Design of Steel Section for Building Frame Openings:

Design Load:

Wind Load Suction: $ULS_s := 2.18kPa$ $SLS_s := 1.42kPa$

Wind Load Pressure: $ULS_p := 1.9kPa$ $SLS_p := 1.24kPa$

$ULS_{wl} := \max(ULS_s, ULS_p) = 2.18 \cdot kPa$ design load ULS

$SLS_{wl} := \max(SLS_s, SLS_p) = 1.42 \cdot kPa$ design load SLS

Loads are based on governing wind load calculation for girt design. Governing WL is for building. Refer to Building Girt Design calculation.

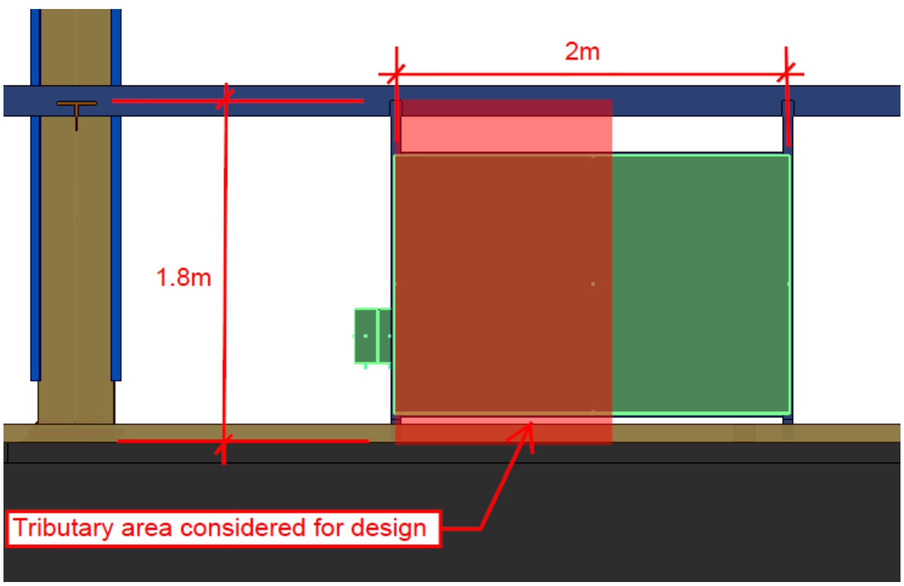
Dimensions:

$s_{girt} := 1.8m$ girt spacing

$s_{column} := 7m$ column spacing

Case 1: For small or large openings that does not interrupt a girt.

Assumptions: 1. For simplicity, design load is based on a full tributary area of the opening.
2. Largest opening width size is 2m.
3. Location of the opening within the column spacing may vary as this does not affect the design since the load is carried by the girts.




Load at steel section: $width_1 := 2m$ width of opening

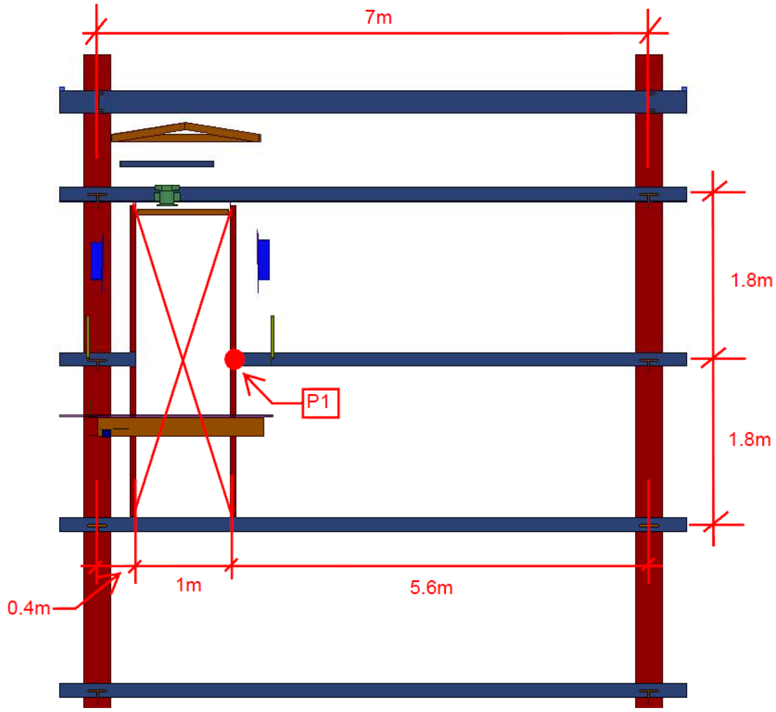
$$load_{case1} := \frac{ULS_{wl} \cdot width_1}{2} = 2.18 \cdot \frac{kN}{m}$$ design load for Case 1

## PURLIN AND GIRT SIZING CALCULATION

Case 2: For door and small/large openings interrupting a single girt.

Assumption: 1. Design load is based on worst case scenario (i.e. door framing is located close to column).

2. Lu = 1.8m due to interrupted girt




Load at steel section:

| | |
|---|---|
| $w_{door} := 1m$ | distance between door frame |
| $door_{col.near} := 0.4m$ | nearest distance from door to column centerline |
| $door_{col.far} := 5.6m$ | farthest distance from door to column centerline |

$$load_{case2.wall} := \frac{ULS_{wl} \cdot door_{col.far} \cdot s_{girt}}{2} = 11 \cdot kN$$

design load due to wall tributary area applied at girt location for Case 2

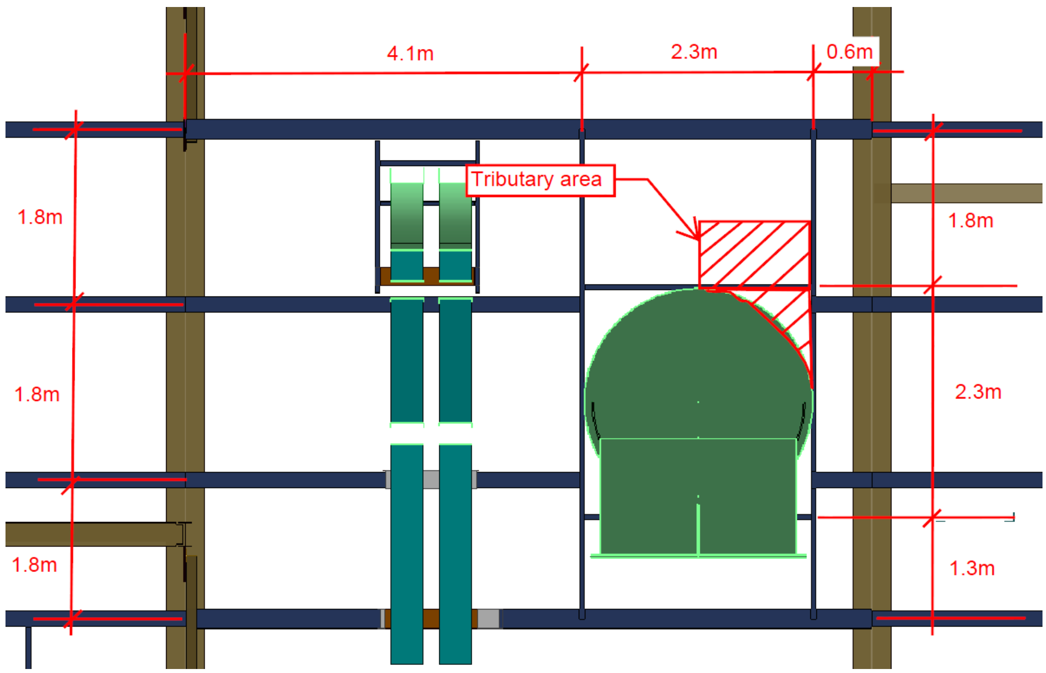
$$load_{case2.door} := \frac{ULS_{wl} \cdot w_{door}}{2} = 1.09 \cdot \frac{kN}{m}$$

design load due to door area applied as a distributed load for Case 2

## PURLIN AND GIRT SIZING CALCULATION

Case 3: This case is applicable to large openings interrupting 2 girts.

Assumption: 1. Design load is based on worst case scenario (i.e. opening is located close to column).
2. Opening is assumed to be square.
3. Maximum location of the opening is at 1.8m.
4. Tributary area at the opening is assumed to be 1/8 of the full tributary area of the opening.
5. Lu = 1.8m due to interrupted girts




Note: Dimensions are not to scale.

Load at steel section: $opening := 2.3m$ width/height of frame opening

$h_{opening} := 1.8m$ maximum distance of frame opening to girt

$opening_{near} := 0.6m$ nearest distance from frame opening to column centerline

$opening_{far} := 4.1m$ farthest distance from frame opening to column centerline

$$load_{case3.wall} := \frac{ULS_{wl} \cdot opening_{far} \cdot s_{girt}}{2} = 8 \cdot kN$$

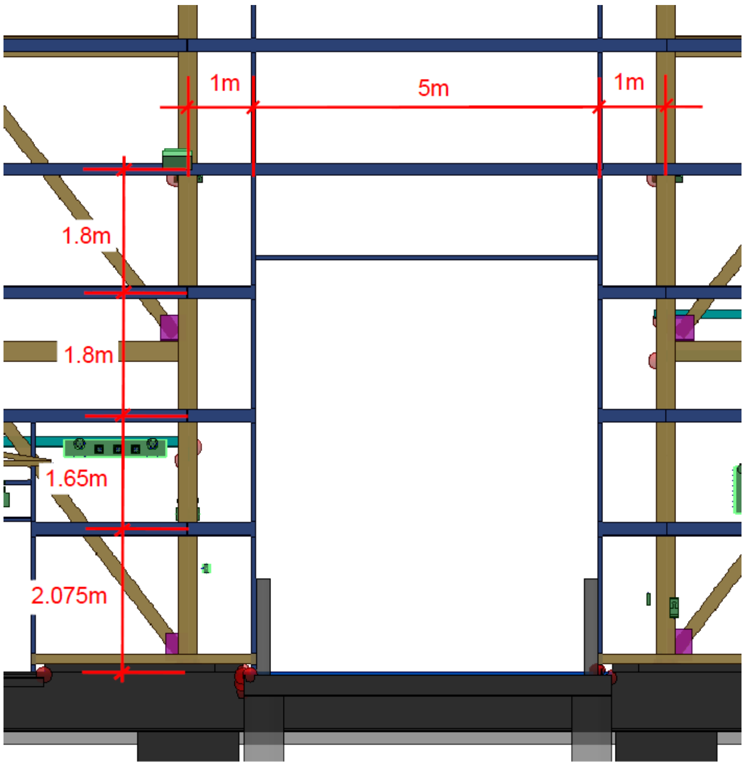
design load due to wall tributray area applied at girt location for Case 3

$$load_{case3.opening} := ULS_{wl} \cdot \left( \frac{opening \cdot h_{opening}}{4} + \frac{opening^2}{8} \right) = 4 \cdot kN$$

design load due to opening tributary area applied at frame opening location for Case 3

**PURLIN AND GIRT SIZING CALCULATION**

Case 4: This case is applicable to overhead door and large openings interrupting 3 girts.
Assumption: 1. Lu = 1.8m due to interrupted girts




Load at steel section: $w_{oh} := 5m$ distance between overhead door frame opening

$opening_{oh} := 1m$ distance from frame opening to column centerline

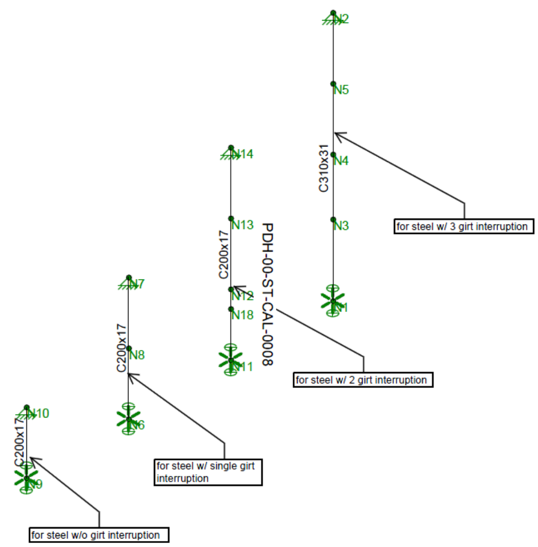
$$load_{case4.wall} := \frac{ULS_{wl} \cdot opening_{oh} \cdot s_{girt}}{2} = 2 \cdot kN$$

$$load_{case4.opening} := \frac{ULS_{wl} \cdot w_{oh}}{2} = 5.45 \cdot \frac{kN}{m}$$

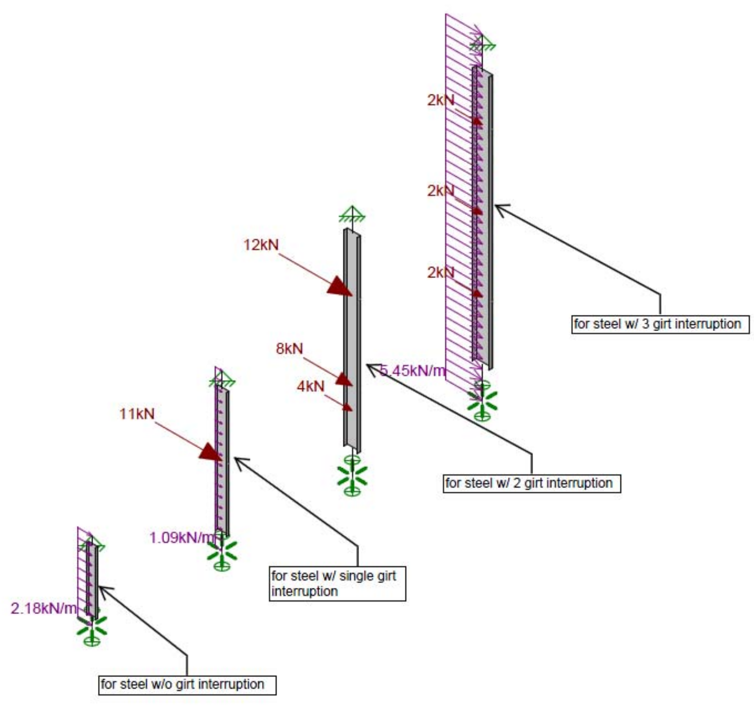
**PURLIN AND GIRT SIZING CALCULATION**

RISA Input and Results:

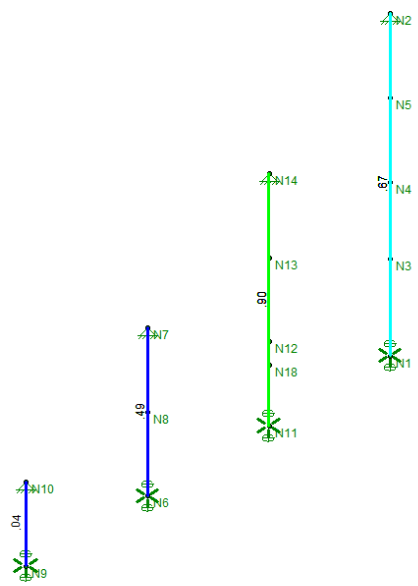
Steel Section Sizes:




Steel Loading:



Steel Unity Check:




Channel section is supported at each girt location. Therefore, model assumption of Lbyy/Lcomp top of 1.8m with pinned boundary condition at each end is conservative.

Note: Deflection result not shown since it is minimal. Refer to RISA model for SLS results.

Summary: Use the following frame steel sizes - C200x17 for no girt interruption
C200x17 for single girt interruption
C200x17 for 2 girt interruption
C310x31 for 3 girt interruption

**PURLIN AND GIRT SIZING CALCULATION**

12. **Check for Channel at Roof Perimeter**

12.1For Stair Tower

Girt Loads: reference - Stair Tower Girt Design

$wt_{wall} := 0.321kPa$ weight of wall

$WL := 2.03kPa$ wind load

$width_{girt} := 0.6m$ tributary width for load distribution - 02HP stair tower governs

Roof Loads: reference - Stair Tower Purlin Design

$wt_{roof} := 0.146kPa$ weight of roof

$SL := 5kPa$ snow load at end channel (refer to snow load calculation)

$width_{roof} := 0.5m$ tributary width for load distribution - 02HP stair tower governs

say Use C250x30

Properties: $\phi_s := 0.9$ $F_y := 300MPa$ $E_s := 200GPa$

$I_x := 32.7\cdot10^6mm^4$ $I_y := 1.16\cdot10^6mm^4$ $S_x := 257\cdot10^3mm^3$ $S_y := 21.5\cdot10^3mm^3$

$L := 6m$ unbraced length

$L_2 := 3m$ assumed unbraced length for channel support about weak axis - conservative

$M_{rx} := 25.5kN\cdot m$ Factored moment resistance about strong axis (for 6m length - CISC Beam Load Table)

$M_{ry} := \phi_s\cdot S_y\cdot F_y = 5.805\ kN\cdot m$ Factored moment resistance about weak axis

Strength Check:

Load case 1: 1.25DL + 1.5SL

$$w_{LC.1} := \left[1.25\cdot\left[\left(wt_{wall}\cdot width_{girt}\right) + \left(wt_{roof}\cdot width_{roof}\right)\right]\right] + 1.5\cdot SL\cdot width_{roof} = 4.082\ \frac{kN}{m}$$

$$Mx_1 := \frac{w_{LC.1}\cdot L^2}{8} = 18.37\ kN\cdot m$$

$$Check_1 := \frac{Mx_1}{M_{rx}} = 0.72$$ less than 1 OK

Load case 2: 1.25 DL + 1.4WL

due to DL $$w_{DL} := \left[1.25\cdot\left[\left(wt_{wall}\cdot width_{girt}\right) + \left(wt_{roof}\cdot width_{roof}\right)\right]\right] = 0.332\ \frac{kN}{m}$$

**PURLIN AND GIRT SIZING CALCULATION**

due to WL $w_{WL} := 1.4 \cdot WL \cdot width_{girt} = 1.71 \frac{kN}{m}$

$$Mx_2 := \frac{w_{DL} \cdot L^2}{8} = 1.49\,kN \cdot m \qquad My_2 := \frac{w_{WL} \cdot L_2^{\,2}}{8} = 1.92\,kN \cdot m$$

$$Check_2 := \frac{Mx_2}{M_{rx}} + \frac{My_2}{M_{ry}} = 0.389$$ less than 1 OK

<u>Deflection Check:</u> assuming no support from roof - conservative

Load Case 1: DL + SL

$$w_{def1} := \left(wt_{wall} \cdot width_{girt}\right) + \left(wt_{roof} \cdot width_{roof}\right) + SL \cdot width_{roof} = 2.77 \frac{kN}{m}$$

$$deflection_1 := \frac{5 \cdot w_{def1} \cdot L^4}{384 \cdot E_s \cdot I_x} = 7.14\,mm$$ less than $\frac{L}{240} = 25\,mm$ OK

Load Case 2: WL

$$deflection_2 := \frac{5 \cdot WL \cdot width_{girt} \cdot L_2^{\,4}}{384 \cdot E_s \cdot I_y} = 5.54\,mm$$ less than $\frac{L_2}{180} = 17\,mm$ OK

## PURLIN AND GIRT SIZING CALCULATION

### Check for Channel at Roof Perimeter

12.2For Building

Girt Loads: reference - Building Girt Design

$wt_{wall} := 0.137kPa$ weight of wall

$WL := 2.19kPa$ wind load

$width_{girt} := 0.7m$ tributary width for load distribution - 01RE building governs

Roof Loads: reference - Building Purlin Design

$wt_{roof} := 0.146kPa$ weight of roof

$SL := 1.59kPa$ snow load at end channel (refer to snow load calculation)

$width_{roof} := 0.375m$ tributary width for load distribution - 01RE building governs

say Use C310x31

Properties: $\phi_s := 0.9$ $F_y := 300MPa$ $E_s := 200GPa$

$I_x := 53.5 \cdot 10^6 mm^4$ $I_y := 1.59 \cdot 10^6 mm^4$ $S_x := 351 \cdot 10^3 mm^3$ $S_y := 28.1 \cdot 10^3 mm^3$

$L := 7m$ unbraced length

$L_2 := 3.5m$ assumed unbraced length for channel support about weak axis - conservative

$M_{rx} := 29.25kN \cdot m$ Factored moment resistance about strong axis
(refer to calculation on page 57)

$M_{ry} := \phi_s \cdot S_y \cdot F_y = 7.587kN \cdot m$ Factored moment resistance about weak axis

Strength Check:

Load case 1: 1.25DL + 1.5SL

$$w_{LC.1} := \left[1.25\left[\left(wt_{wall} \cdot width_{girt}\right) + \left(wt_{roof} \cdot width_{roof}\right)\right]\right] + 1.5 \cdot SL \cdot width_{roof} = 1.083\frac{kN}{m}$$

$$Mx_1 := \frac{w_{LC.1} \cdot L^2}{8} = 6.63kN \cdot m$$

$$Check_1 := \frac{Mx_1}{M_{rx}} = 0.227$$ less than 1 OK

Load case 2: 1.25 DL + 1.4WL

due to DL $$w_{DL} := \left[1.25 \cdot \left[\left(wt_{wall} \cdot width_{girt}\right) + \left(wt_{roof} \cdot width_{roof}\right)\right]\right] = 0.188\frac{kN}{m}$$

**PURLIN AND GIRT SIZING CALCULATION**

due to WL $w_{WL} := 1.4 \cdot WL \cdot width_{girt} = 2.15 \frac{kN}{m}$

$Mx_2 := \frac{w_{DL} \cdot L^2}{8} = 1.15\,kN \cdot m$ $My_2 := \frac{w_{WL} \cdot L_2^{\,2}}{8} = 3.29\,kN \cdot m$

$Check_2 := \frac{Mx_2}{M_{rx}} + \frac{My_2}{M_{ry}} = 0.473$ less than 1 OK

Deflection Check: assuming no support from roof - conservative

Load Case 1: DL + SL

$$w_{def1} := \left(wt_{wall} \cdot width_{girt}\right) + \left(wt_{roof} \cdot width_{roof}\right) + SL \cdot width_{roof} = 0.75 \frac{kN}{m}$$

$deflection_1 := \frac{5 \cdot w_{def1} \cdot L^4}{384 \cdot E_s \cdot I_x} = 2.18\,mm$ less than $\frac{L}{240} = 29.167\,mm$ OK

Load Case 2: WL

$deflection_2 := \frac{5 \cdot WL \cdot width_{girt} \cdot L_2^{\,4}}{384 \cdot E_s \cdot I_y} = 9.42\,mm$ less than $\frac{L_2}{180} = 19\,mm$ OK

## PURLIN AND GIRT SIZING CALCULATION

Properties of C310x31

| | |
|---|---|
| $b_c := 74mm$ | flange width |
| $d_c := 305mm$ | depth |
| $t_{fl.c} := 12.7mm$ | flange thickness |
| $t_{w.c} := 7.2mm$ | web thickness |
| $A_c := 3920mm^2$ | Area |
| $I_{x.c} := 53.5 \cdot 10^6 mm^4$ | Moment of Inertia(Strong Axis) |
| $S_{x.c} := 351 \cdot 10^3 mm^3$ | Elastic Section Modulus(Strong Axis) |
| $I_{y.c} := 1.59 \cdot 10^6 mm^4$ | Moment of Inertia(Weak Axis) |
| $S_{y.c} := 28.1 \cdot 10^3 mm^3$ | Elastic Section Modulus(Weak Axis) |
| $J_c := 152 \cdot 10^3 \cdot mm^4$ | Torsional Constant |
| $C_{w.c} := 29.3 \cdot 10^9 \cdot mm^6$ | Warping Constant |
| $F_{y.c} := 300MPa$ | Yield Strength |

Properties of Steel:

$E_s := 200GPa$ $G_s := 77GPa$ $\phi_s := 0.9$

$L := 7m$ unbraced length

Load: $w := 1\frac{kN}{m}$

$M_{max} := \frac{w \cdot L^2}{8} = 6.125kN \cdot m$ maximum moment

$M_a := \frac{3w \cdot L^2}{32} = 4.594kN \cdot m$ moment at 1/4 point

$M_b := M_{max} = 6.125\ kN \cdot m$ moment at midpoint

$M_c := \frac{3w \cdot L^2}{32} = 4.594\ kN \cdot m$ moment at 3/4 point

Channel Moment Capacity: Cl. 13.6.b

$$\omega_2 := \min\left(2.5, \frac{4 \cdot M_{max}}{\sqrt{M_{max}^2 + 4 \cdot M_a^2 + 7 \cdot M_b^2 + 4 \cdot M_c^2}}\right) = 1.131$$

$$M_u := \frac{\omega_2 \cdot \pi}{L} \cdot \sqrt{E_s \cdot I_{y.c} \cdot G_s \cdot J_c + \left(\frac{\pi \cdot E_s}{L}\right)^2 \cdot I_{y.c} \cdot C_{w.c}} = 32.5 \cdot kN \cdot m$$ critical elastic moment

$$M_{rx} := \begin{cases} \min\left[1.15 \cdot \phi_s \cdot S_{x.c} \cdot F_{y.c} \cdot \left(1 - \frac{0.28 \cdot S_{x.c} \cdot F_{y.c}}{M_u}\right), \phi_s \cdot S_{x.c} \cdot F_{y.c}\right] & \text{if } M_u > 0.67 \cdot S_{x.c} \cdot F_{y.c} \\ \phi_s \cdot M_u & \text{otherwise} \end{cases}$$

$M_{rx} = 29.25 \cdot kN \cdot m$ Channel moment capacity

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SHEET NO.

PDH-00-ST-CAL-0008

Purlin and Sizing Calculation

Check Purlin located close to channel edge

for Stair Tower : Purlin Section = W200X36 (assuming support at midpoint – 3m)

$WL = 1.4\,(2.03\,kPa)(0.6m)(3m) = 5.11\,kN$

$My_{max} = 5.11kN(6m)/4 = 7.7\,kN\text{-}m$ ⇒ due to WL on Wall

$Mry = 0.9 Zy Sy = 0.9(141 \times 10^3 mm^3)(345MPa) = 43.78\,kN\text{-}m$

$$Mx = \frac{1.4(3.58kPa)(0.5m + 0.35m)(6m)^2}{8} = 19.2\,kN\text{-}m$$ ⇒ due to WL on Roof (refer to ST purlin design for design load)

$Mrx = 71.3\,kN\text{-}m$ for 6m length – CISC beam load table

$$UC = \frac{Mx}{Mrx} + \frac{My_{max}}{My} = \frac{19.2\,kN\text{-}m}{71.3kN} + \frac{7.7kN\text{-}m}{43.78\,kN\text{-}m} = 0.45 < 1 \text{ OK}$$

for Building : Purlin Section = W310X39 (assuming support at midpoint – 3.5m)

$WL = 1.4(2.19\,kPa)(0.7m)(3.5m) = 7.5\,kN$

$My_{max} = 7.5kN(7m)/4 = 13.13\,kN\text{-}m$ ⇒ due to WL on Wall

$Mry = 0.9 Zy Sy = 0.9(135 \times 10^3 mm^3)(345MPa) = 41.9\,kN\text{-}m$

$$Mx = \frac{1.4(3.84kPa)(0.375m + 0.75m)(7m)^2}{8} = 37\,kN\text{-}m$$ ⇒ due to WL on Roof (refer to building purlin design for design load)

$Mrx = 62.2\,kN\text{-}m$ for 7m length – CISC beam load table

$$UC = \frac{Mx}{Mrx} + \frac{My_{max}}{My} = \frac{37\,kN\text{-}m}{62.2kN\text{-}m} + \frac{13.13\,kN\text{-}m}{41.9\,kN\text{-}m} = 0.91 < 1 \text{ OK}$$

Appendix A: AWIP Roof/Wall Information

ALL WEATHER INSULATED PANELS

## 40'' Wide Exterior Wall Panel Allowable Loads (PSF)

| Panel Thickness | Span Condition | Fastening Pattern | Panel Span (ft) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
| 2'' | Single Span | FS-A | 48 | 40 | 34 | 30 | 25 | 20 | 16 | 13 | 11 |
| | | FS-B | 56 | 47 | 40 | 32 | 25 | 20 | 16 | 13 | 11 |
| | | FS-C | 56 | 47 | 40 | 32 | 25 | 20 | 16 | 13 | 11 |
| | | FS-D | 56 | 47 | 40 | 32 | 25 | 20 | 16 | 13 | 11 |
| | | FS-E | 56 | 47 | 40 | 32 | 25 | 20 | 16 | 13 | 11 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 11 |
| | Two Spans | FS-A | 48 | 40 | 34 | 29 | 26 | 23 | 21 | 19 | 16 |
| | | FS-B | 50 | 41 | 34 | 29 | 26 | 23 | 21 | 19 | 16 |
| | | FS-C | 50 | 41 | 34 | 29 | 26 | 23 | 21 | 19 | 16 |
| | | FS-D | 50 | 41 | 34 | 29 | 26 | 23 | 21 | 19 | 16 |
| | | FS-E | 50 | 41 | 34 | 29 | 26 | 23 | 21 | 19 | 16 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| Weight = 2.22 PSF | Three or More Spans | FS-A | 48 | 40 | 34 | 30 | 26 | 23 | 21 | 19 | 16 |
| | | FS-B | 49 | 40 | 34 | 30 | 26 | 23 | 21 | 19 | 16 |
| | | FS-C | 49 | 40 | 34 | 30 | 26 | 23 | 21 | 19 | 16 |
| | | FS-D | 49 | 40 | 34 | 30 | 26 | 23 | 21 | 19 | 16 |
| | | FS-E | 49 | 40 | 34 | 30 | 26 | 23 | 21 | 19 | 16 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| 2.5'' | Single Span | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 19 | 16 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 29 | 23 | 19 | 16 |
| | | FS-C | 70 | 59 | 50 | 44 | 35 | 29 | 23 | 19 | 16 |
| | | FS-D | 70 | 59 | 50 | 44 | 35 | 29 | 23 | 19 | 16 |
| | | FS-E | 70 | 59 | 50 | 44 | 35 | 29 | 23 | 19 | 16 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| | Two Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 37 | 33 | 29 | 26 | 24 | 22 |
| | | FS-C | 63 | 52 | 43 | 37 | 33 | 29 | 26 | 24 | 22 |
| | | FS-D | 63 | 52 | 43 | 37 | 33 | 29 | 26 | 24 | 22 |
| | | FS-E | 63 | 52 | 43 | 37 | 33 | 29 | 26 | 24 | 22 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| Weight = 2.34 PSF | Three or More Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 37 | 33 | 30 | 27 | 24 | 22 |
| | | FS-C | 62 | 51 | 43 | 37 | 33 | 30 | 27 | 24 | 22 |
| | | FS-D | 62 | 51 | 43 | 37 | 33 | 30 | 27 | 24 | 22 |
| | | FS-E | 62 | 51 | 43 | 37 | 33 | 30 | 27 | 24 | 22 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| 3'' | Single Span | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 22 |
| | | FS-C | 85 | 71 | 60 | 53 | 46 | 38 | 31 | 26 | 22 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 31 | 26 | 22 |
| | | FS-E | 85 | 71 | 60 | 53 | 46 | 38 | 31 | 26 | 22 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| | Two Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 77 | 63 | 53 | 46 | 40 | 36 | 32 | 29 | 27 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 32 | 29 | 27 |
| | | FS-E | 77 | 63 | 53 | 46 | 40 | 36 | 32 | 29 | 27 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| Weight = 2.41 PSF | Three or More Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 75 | 62 | 52 | 45 | 40 | 36 | 32 | 30 | 27 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 32 | 30 | 27 |
| | | FS-E | 75 | 62 | 52 | 45 | 40 | 36 | 32 | 30 | 27 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |

| Panel Thickness | Span Condition | Fastening Pattern | Panel Span (ft) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
| 4'' | Single Span | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 37 | 32 | 28 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 52 | 44 | 37 | 32 | 28 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| | Two Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 43 | 38 | 34 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| Weight = 2.62 PSF | Three or More Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 43 | 38 | 34 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| 5'' | Single Span | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 44 | 40 | 37 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| | Two Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 44 | 40 | 37 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| Weight = 2.82 PSF | Three or More Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 44 | 40 | 37 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| 6'' | Single Span | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 44 | 40 | 37 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| | Two Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 44 | 40 | 37 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |
| Weight = 2.98 PSF | Three or More Spans | FS-A | 48 | 40 | 34 | 30 | 27 | 24 | 22 | 20 | 18 |
| | | FS-B | 60 | 50 | 43 | 38 | 33 | 30 | 27 | 25 | 23 |
| | | FS-C | 85 | 71 | 61 | 53 | 47 | 42 | 38 | 35 | 32 |
| | | FS-D | 73 | 61 | 52 | 45 | 40 | 36 | 33 | 30 | 28 |
| | | FS-E | 97 | 81 | 69 | 61 | 54 | 48 | 44 | 40 | 37 |
| | | FS-F | 32 | 27 | 23 | 20 | 18 | 16 | 14 | 13 | 12 |

Notes:

1. Spans shown are based on transverse load testing per ASTM-E72 and strength of fastening patterns. Fastening is based on Tek 3 fasteners installed on minimum 16 gauge girts.
2. Spans calculated with 26 gauge exterior and interior facings.
3. Deflection Limit: L/180
4. Safety factor = 2.5 for buckling, 3.0 for shear, 3.0 for fastening
5. FS-A = WC-01 Wall Clip with (3) fasteners, FS-F = WC-01 with (2) fasteners
6. FS-B = WC-01 with (3) fasteners + (1) Fablok
7. FS-C = WC-01 with (3) fasteners + (3) Fablok
8. FS-D = WC-01 with (3) fasteners + (2) Fablok
9. FS-E = WC-01 with (3) fasteners + (4) Fablok
10. Thermal effect due to temperature differentials have not been considered.
11. For FiRe Panel, use load-span values for 4'' wall. FiRe Panel weight is 5.5 PSF.
12. Consult your AWIP/Vicwest representative for project specific requirements.
13. Consult your AWIP/Vicwest representative for FM Global Loss Prevention Data Sheet 1-28 requirements.

Rev: July 01, 2015

ALL WEATHER INSULATED PANELS

Stair Tower Purlin Span

Building Purlin Span

# SR2 Panel Allowable Loads (PSF)

| Panel Thickness | Panel Weight | Design Criteria | Panel Span (ft) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 2'-6'' | 3'-0'' | 3'-6'' | 4'-0'' | 4'-6'' | 5'-0'' | 5'-6'' | 6'-0'' | 7'-0'' |
| 3.25'' | 2.48 | Panel/Deflection Limit | 149 | 123 | 104 | 90 | 79 | 70 | 62 | 56 | 46 |
| 4'' | 2.65 | Panel/Deflection Limit | 185 | 153 | 130 | 112 | 98 | 87 | 78 | 71 | 59 |
| 5'' | 2.86 | Panel/Deflection Limit | 233 | 192 | 164 | 142 | 125 | 111 | 100 | 90 | 75 |
| 6'' | 3.12 | Panel/Deflection Limit | 262 | 233 | 198 | 172 | 151 | 135 | 121 | 110 | 92 |

| Fastening Pattern | Connection Strength | Panel Span (ft) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | 2'-6'' | 3'-0'' | 3'-6'' | 4'-0'' | 4'-6'' | 5'-0'' | 5'-6'' | 6'-0'' | 7'-0'' |
| (2) Fasteners per clip | 16 ga purlins | 70 | 59 | 50 | 44 | 39 | 35 | 32 | 29 | 25 |
| | 12 ga purlins | 149 | 124 | 106 | 93 | 83 | 75 | 68 | 62 | 53 |
| | 3/16'' thick purlins | ***160*** | ***134*** | ***116*** | ***101*** | ***90*** | ***81*** | ***74*** | ***67*** | ***58*** |
| (3) Fasteners per clip | 16 ga purlins | 105 | 88 | 76 | 66 | 59 | 53 | 48 | 44 | 38 |
| | 12 ga purlins | ***160*** | ***134*** | ***116*** | ***101*** | ***90*** | ***81*** | ***74*** | ***67*** | ***58*** |

Notes:

1. Spans shown are based on transverse load testing per ASTM-E72 and strength of fastening patterns.
2. Spans calculated with 26 gauge exterior and interior facings.
3. The lowest allowable load between panel design and connection strength must be used to determine maximum span.
4. Fastening calculated with 1/4-14 Tek 3 for 16 gauge and 12 gauge purlins and 1/4-20 Tek 5 for 3/16'' thick purlins.
5. Deflection Limit: L/240
6. Safety factor = 2.5 for buckling, 3.0 for shear, 3.0 for fastening
7. Allowable suction loads in bold-italic font may be greater with the use of the EC-01 Enhanced Standing Seam Clip.
8. Structural capacity of purlins have not been considered.
9. Thermal effect due to temperature differentials have not been considered.
10. Consult your AWIP/Vicwest representative for project specific requirements.
11. Consult your AWIP/Vicwest representative for FM Global Loss Prevention Data Sheet 1-28 requirements.

Rev: July 1, 2015

Weight Table

# KS Series

| | WEIGHT PER SQ. FT. | | |
|---|---|---|---|
| | 26 GA / 26 GA | 24 GA / 26 GA | 24 GA / 24 GA |
| PANEL THICKNESS | | | |
| 2” | 2.30 | 2.49 | 2.66 |
| | | | |
| 2.5” | 2.41 | 2.60 | 2.78 |
| | | | |
| 3” | 2.50 | 2.69 | 2.86 |
| | | | |
| 4” | 2.70 | 2.89 | 3.06 |
| | | | |
| 5” | 2.89 | 3.08 | 3.26 |
| | | | |
| 6” | 3.08 | 3.26 | 3.44 |

| | WEIGHT PER SQ. FT. | | |
|---|---|---|---|
| | 22 GA / 26 GA | 22 GA / 24 GA | 22 GA / 22 GA |
| PANEL THICKNESS | | | |
| 2” | 2.77 | 2.94 | 3.21 |
| | | | |
| 2.5” | 2.88 | 3.06 | 3.32 |
| | | | |
| 3” | 2.97 | 3.14 | 3.41 |
| | | | |
| 4” | 3.17 | 3.34 | 3.61 |
| | | | |
| 5” | 3.36 | 3.53 | 3.80 |
| | | | |
| 6” | 3.54 | 3.72 | 3.98 |

Note: KS Series panels include KS Optimo, KS42SL, KS42MR, KS42MMR, KS42MW and KS45SL. Refer to other weight tables for KS Granitstone and KS Mineral Fiber panels.

| PANEL THICKNESS | SPAN CONDITION | NEGATIVE WALL PRESSURE IN PSF | | | | | |
|---|---|---|---|---|---|---|---|
| | | 20 PSF | 25 PSF | 30 PSF | 35 PSF | 40 PSF | 45 PSF |
| 2 | double span | 11'-2" | 9'-1" | 7'-8" | 6'-8" | 5'-11" | 5'-3" |
| | multiple span | 11'-4" | 9'-2" | 7'-8" | 6'-7" | 5'-10" | 5'-3" |
| 2.5 | double span | 14'-1" | 11'-5" | 9'-8" | 8'-5" | 7'-5" | 6'-8" |
| | multiple span | 14'-0" | 11'-7" | 9'-9" | 8'-5" | 7'-5" | 6'-7" |
| 3 | double span | 15'-9" | 12'-9" | 10'-9" | 9'-4" | 8'-3" | 7'-5" |
| | multiple span | 16'-3" | 14'-0" | 11'-9" | 10'-1" | 8'-10" | 7'-11" |
| 4 | double span | 15'-11" | 12'-11" | 10'-11" | 9'-6" | 8'-5" | 7'-7" |
| | multiple span | 17'-6" | 14'-1" | 11'-9" | 10'-1" | 8'-11" | 7'-11" |
| 5 | double span | 16'-2" | 13'-2" | 11'-2" | 9'-8" | 8'-7" | 7'-9" |
| | multiple span | 17'-7" | 14'-1" | 11'-10" | 10'-2" | 8'-11" | 8'-0" |
| 6 | double span | 16'-3" | 13'-3" | 11'-3" | 9'-10" | 8'-8" | 7'-10" |
| | multiple span | 17'-7" | 14'-2" | 11'-10" | 10'-3" | 9'-0" | 8'-0" |

Panel Series: KS Shadowline
Design Parameter: Uniform Load
Fastening Method: Standard 2-Hole Concealed Clip w/ (2) 1/4" TEK Fasteners
Exterior Face: 26 Ga. Galvanized Steel
Interior Face: 26 Ga. Galvanized Steel
Deflection Limit: L/180
Structural Support Gauge: 14 Ga
Panel Width: 24"
Thermal Differential: Not taken into account

| PANEL THICKNESS | SPAN CONDITION | NEGATIVE WALL PRESSURE IN PSF | | | | | |
|---|---|---|---|---|---|---|---|
| | | 20 PSF | 25 PSF | 30 PSF | 35 PSF | 40 PSF | 45 PSF |
| 2 | double span | 11'-2" | 9'-1" | 7'-8" | 6'-8" | 5'-11" | 5'-3" |
| | multiple span | 11'-4" | 9'-2" | 7'-8" | 6'-7" | 5'-10" | 5'-3" |
| 2.5 | double span | 12'-8" | 10'-4" | 8'-9" | 7'-7" | 6'-9" | 6'-0" |
| | multiple span | 14'-0" | 11'-3" | 9'-5" | 8'-1" | 7'-1" | 6'-4" |
| 3 | double span | 12'-9" | 10'-5" | 8'-9" | 7'-8" | 6'-9" | 6'-1" |
| | multiple span | 14'-0" | 11'-3" | 9'-5" | 8'-1" | 7'-1" | 6'-4" |
| 4 | double span | 12'-11" | 10'-6" | 8'-11" | 7'-9" | 6'-11" | 6'-2" |
| | multiple span | 14'-1" | 11'-3" | 9'-5" | 8'-2" | 7'-2" | 6'-5" |
| 5 | double span | 13'-2" | 10'-9" | 9'-1" | 7'-11" | 7'-0" | 6'-4" |
| | multiple span | 14'-1" | 11'-4" | 9'-6" | 8'-3" | 7'-3" | 6'-6" |
| 6 | double span | 13'-3" | 10'-10" | 9'-3" | 8'-0" | 7'-1" | 6'-5" |
| | multiple span | 14'-2" | 11'-5" | 9'-7" | 8'-3" | 7'-3" | 6'-6" |

Panel Series: KS Shadowline
Design Parameter: Uniform Load
Fastening Method: Standard 2-Hole Concealed Clip w/ (2) 1/4" TEK Fasteners
Exterior Face: 26 Ga. Galvanized Steel
Interior Face: 26 Ga. Galvanized Steel
Deflection Limit: L/180
Structural Support Gauge: 14 Ga
Panel Width: 30"
Thermal Differential: Not taken into account

| PANEL THICKNESS | SPAN CONDITION | NEGATIVE WALL PRESSURE IN PSF | | | | | |
|---|---|---|---|---|---|---|---|
| | | 20 PSF | 25 PSF | 30 PSF | 35 PSF | 40 PSF | 45 PSF |
| 2 | double span | 10'-7" | 8'-7" | 7'-3" | 6'-4" | 5'-7" | 5'-1" |
| | multiple span | 11'-4" | 9'-2" | 7'-8" | 6'-7" | 5'-10" | 5'-3" |
| 2.5 | double span | 10'-9" | 8'-9" | 7'-5" | 6'-5" | 5'-8" | 5'-1" |
| | multiple span | 11'-8" | 9'-5" | 7'-10" | 6'-9" | 5'-11" | 5'-4" |
| 3 | double span | 10'-9" | 8'-9" | 7'-5" | 6'-6" | 5'-9" | 5'-2" |
| | multiple span | 11'-9" | 9'-5" | 7'-11" | 6'-10" | 6'-0" | 5'-4" |
| 4 | double span | 10'-11" | 8'-11" | 7'-7" | 6'-7" | 5'-10" | 5'-2" |
| | multiple span | 11'-9" | 9'-5" | 7'-11" | 6'-10" | 6'-0" | 5'-4" |
| 5 | double span | 11'-2" | 9'-1" | 7'-9" | 6'-9" | 6'-0" | 5'-4" |
| | multiple span | 11'-10" | 9'-6" | 8'-0" | 6'-11" | 6'-1" | 5'-5" |
| 6 | double span | 11'-3" | 9'-3" | 7'-10" | 6'-9" | 6'-0" | 5'-5" |
| | multiple span | 11'-10" | 9'-7" | 8'-0" | 6'-11" | 6'-1" | 5'-5" |

Panel Series: KS Shadowline
Design Parameter: Uniform Load
Fastening Method: Standard 2-Hole Concealed Clip w/ (2) 1/4" TEK Fasteners
Exterior Face: 26 Ga. Galvanized Steel
Interior Face: 26 Ga. Galvanized Steel
Deflection Limit: L/180
Structural Support Gauge: 14 Ga
Panel Width: 36"
Thermal Differential: Not taken into account

| PANEL THICKNESS | SPAN CONDITION | NEGATIVE WALL PRESSURE IN PSF | | | | | |
|---|---|---|---|---|---|---|---|
| | | 20 PSF | 25 PSF | 30 PSF | 35 PSF | 40 PSF | 45 PSF |
| 2 | double span | 9'-2" | 7'-6" | 6'-4" | 5'-6" | 4'-11" | 4'-5" |
| | multiple span | 10'-0" | 8'-1" | 6'-9" | 5'-10" | 5'-1" | 4'-7" |
| 2.5 | double span | 9'-3" | 7'-7" | 6'-5" | 5'-7" | 5'-0" | 4'-4" |
| | multiple span | 10'-1" | 8'-1" | 6'-9" | 5'-10" | 5'-2" | 4'-6" |
| 3 | double span | 9'-4" | 7'-8" | 6'-6" | 5'-8" | 5'-0" | 4'-4" |
| | multiple span | 10'-1" | 8'-1" | 6'-10" | 5'-10" | 5'-2" | 4'-6" |
| 4 | double span | 9'-6" | 7'-9" | 6'-7" | 5'-9" | 4'-11" | 4'-4" |
| | multiple span | 10'-1" | 8'-2" | 6'-10" | 5'-11" | 5'-2" | 4'-6" |
| 5 | double span | 9'-8" | 7'-11" | 6'-9" | 5'-10" | 5'-1" | 4'-6" |
| | multiple span | 10'-2" | 8'-3" | 6'-11" | 5'-11" | 5'-3" | 4'-7" |
| 6 | double span | 9'-10" | 8'-0" | 6'-9" | 5'-11" | 5'-3" | 4'-8" |
| | multiple span | 10'-3" | 8'-3" | 6'-11" | 6'-0" | 5'-3" | 4'-8" |

Panel Series: KS Shadowline
Design Parameter: Uniform Load
Fastening Method: Standard 2-Hole Concealed Clip w/ (2) 1/4" TEK Fasteners
Exterior Face: 26 Ga. Galvanized Steel
Interior Face: 26 Ga. Galvanized Steel
Deflection Limit: L/180
Structural Support Gauge: 14 Ga
Panel Width: 42"
Thermal Differential: Not taken into account

NOTES TO TABLE:
Spans shown are for negative uniform loads only. Spans may be governed by other factors including temperature differentials, deflection requirements, structural support gage, or unequal spacing of secondary. Consult Kingspan for specific project applications.

# KS42 MF

| | WALL PANEL SPAN IN FEET (L/180) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 5 | 10 | 12 | 14 | 16 | 18 | 20 | 25 |
| **PANEL THICKNESS** | DESIGN LOAD IN PSF | | | | | | | |
| 4” / 6” / 8” | 63 | 32 | 26 | 23 | 20 | 17 | 14 | 8 |

## CONDITIONS:

Panel Series: KS42 MF

Design Parameter: Uniform Load

Fastening Method: Expansion Fastener

Exterior Face: 26 Ga. Galvanized Steel

Interior Face: 26 Ga. Galvanized Steel

Deflection Limit: L/180 or 1” maximum

## NOTES TO TABLE:

1) Spans shown are simple and limited by deflection under uniform load only. Spans may be governed by other factors including deflection due to temperature differentials, multiple spans, negative wind pressure and connection points required for fasteners. Consult Kingspan for specific project applications.

| | WEIGHT PER SQ. FT. | | |
|---|---|---|---|
| | 26 GA / 26 GA | 24 GA / 26 GA | 24 GA / 24 GA |
| **PANEL THICKNESS** | | | |
| 4” | 4.80 | 5.00 | 5.20 |
| 6” | 6.30 | 6.50 | 6.70 |
| 8” | 7.80 | 8.00 | 8.20 |



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| ALLOWABLE POSITIVE AND NEGATIVE WIND LOADS (PSF) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PANEL THICKNESS | SPAN CONDITION | DESIGN CRITERIA | SUPPORT SPAN (FEET) | | | | | | | | | | |
| | | | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 4” | two or more spans | buckling | 86 | 67 | 54 | 46 | 42 | 39 | 35 | 32 | 29 | 27 | 25 |
| | | deflection (L/240) | 116 | 93 | 76 | 64 | 55 | 47 | 41 | 36 | 32 | 28 | 25 |
| | | connection* | 48.8 | 40.6 | 34.7 | 30.5 | 27.1 | 24.4 | 22.2 | 20.3 | 18.7 | 17.4 | 16.2 |
| | | connection** | 65.1 | 54.2 | 46.5 | 40.7 | 36.1 | 32.5 | 29.6 | 27.1 | 25 | 23.2 | 21.7 |
| 5” | two or more spans | buckling | 108 | 85 | 73 | 66 | 61 | 55 | 51 | 46 | 42 | 39 | 35 |
| | | deflection (L/240) | 147 | 119 | 98 | 83 | 71 | 62 | 54 | 48 | 42 | 38 | 34 |
| | | connection* | 49.4 | 41.2 | 35.3 | 30.9 | 27.4 | 24.7 | 22.4 | 20.6 | 19 | 17.6 | 16.4 |
| | | connection** | 65.7 | 54.8 | 46.9 | 41.1 | 36.5 | 32.8 | 29.8 | 27.4 | 25.2 | 23.4 | 21.9 |
| 6” | two or more spans | buckling | 110 | 88 | 79 | 73 | 68 | 62 | 57 | 52 | 47 | 43 | 39 |
| | | deflection (L/240) | 179 | 145 | 120 | 102 | 88 | 77 | 67 | 60 | 53 | 48 | 43 |
| | | connection* | 50 | 41.6 | 35.7 | 31.2 | 27.7 | 25 | 22.7 | 20.8 | 19.2 | 17.8 | 16.6 |
| | | connection** | 66.2 | 55.1 | 47.3 | 41.4 | 36.8 | 33.1 | 30.1 | 27.6 | 25.4 | 23.6 | 22 |
| 8” | two or more spans | buckling | 141 | 115 | 109 | 102 | 95 | 87 | 80 | 73 | 67 | 61 | 55 |
| | | deflection (L/240) | 243 | 198 | 165 | 141 | 122 | 107 | 94 | 84 | 76 | 68 | 62 |
| | | connection* | 51.1 | 42.6 | 36.5 | 31.9 | 28.4 | 25.5 | 23.2 | 21.3 | 19.6 | 18.2 | 17 |
| | | connection** | 67.3 | 56.1 | 48.1 | 42.1 | 37.4 | 33.7 | 30.6 | 28.1 | 25.9 | 24.1 | 22.4 |

## NOTES TO TABLE:

1) Table applies to KS42MF Through Fix panel with 26/26 Ga. exterior & interior steel facing. Steel Grade A with minimum Fy = 33 ksi.
2) Allowable positive load is the lowest value of panel buckling stress & deflection limit.
3) Allowable negative load is the lowest value of panel buckling stress, deflection limit and connection strength for each fastener pattern.
4) Connection* loads are based on 3 x #14 type B self tapping fasteners c/w 1” dia washer.
5) Connection** loads are based on loads are based on 4 x #14 type B self tapping fasteners c/w 1” dia washer.
6) The fastening capacities were derived from either fastener pull out or pull over testing.
7) The structural capacity of the steel support members is sole responsibility of a licensed structural engineer.
8) Thermal loading has not been considered.

| | WEIGHT PER SQ. FT. | | |
|---|---|---|---|
| | 26 GA / 26 GA | 26 GA / 24 GA | 24 GA / 24 GA |
| PANEL THICKNESS | | | |
| 4” | 4.80 | 5.00 | 5.20 |
| 6” | 6.29 | 6.55 | 2.83 |

# 4" KingZip 24/24 - 42"
## Wind Load Analysis

KINGSPAN

"Short Term" Wind load

Stair Tower Purlin Span

## Allowable Load for Panel

| Span | Single Span | | Double Span | | Triple Span | |
|---|---|---|---|---|---|---|
| Ft - In | Load PSF | Governing Factor | Load PSF | Governing Factor | Load PSF | Governing Factor |
| 2' - 0" | 158.43 | End Connection | 160.33 | End Connection | 162.02 | End Connection |
| 2' - 6" | 126.74 | End Connection | 129.08 | End Connection | 131.02 | End Connection |
| 3' - 0" | 105.62 | End Connection | 108.35 | End Connection | 110.46 | End Connection |
| 3' - 6" | 90.53 | End Connection | 93.63 | End Connection | 95.83 | End Connection |
| 4' - 0" | 79.21 | End Connection | 82.65 | End Connection | 84.86 | End Connection |
| 4' - 6" | 70.41 | End Connection | 74.15 | End Connection | 76.32 | End Connection |
| 5' - 0" | 63.37 | End Connection | 67.38 | End Connection | 69.46 | End Connection |
| 5' - 6" | 57.61 | End Connection | 61.85 | End Connection | 63.80 | End Connection |
| 6' - 0" | 52.81 | End Connection | 57.24 | End Connection | 59.06 | End Connection |
| 6' - 6" | 48.75 | End Connection | 53.35 | End Connection | 55.00 | End Connection |
| [illegible] | 45.27 | End Connection | 50.01 | End Connection | 51.49 | End Connection |
| [illegible] | 42.25 | End Connection | 46.66 | Int Connection | 48.39 | Int Connection |
| 8' - 0" | 39.61 | End Connection | 43.43 | Int Connection | 45.24 | Int Connection |
| 8' - 6" | 37.28 | End Connection | 40.60 | Int Connection | 42.47 | Int Connection |
| 9' - 0" | 35.21 | End Connection | 38.10 | Int Connection | 40.02 | Int Connection |
| 9' - 6" | 33.35 | End Connection | 35.88 | Int Connection | 37.84 | Int Connection |
| 10' - 0" | 31.69 | End Connection | 33.90 | Int Connection | 35.88 | Int Connection |
| 10' - 6" | 30.18 | End Connection | 32.11 | Int Connection | 34.12 | Int Connection |
| 11' - 0" | 28.81 | End Connection | 30.50 | Int Connection | 32.52 | Int Connection |
| 11' - 6" | 27.55 | End Connection | 29.04 | Int Connection | 31.07 | Int Connection |
| 12' - 0" | 26.40 | End Connection | 27.72 | Int Connection | 29.73 | Int Connection |

Building Purlin Span

NOTES:

1) Section properties, ultimate values and safety factors used to generate this table:

$I_c = 2.7370$ $In^4/Ft$ $S_{bf} = 0.7512$ $In^3/Ft$ $S_{bl} = 1.2834$ $In^3/Ft$

$A_c = 48.45$ $In^2/Ft$ $G_c = 433$ psi E = 29,500,000 psi

$F_b = 24852$ psi w/ Safety Factor = 2.50 $F_v = 17.80$ psi w/ Safety Factor = 3.00

End Connection = 1109 Lb w/ Safety Factor = 2.00 Int Connection = 2702 Lb w/ Safety Factor = 2.00

End Supt. P.O. = 6906 Lb w/ Safety Factor = 3.00 Interior Supt. P.O. = 6906 Lb w/ Safety Factor = 3.00

2) An allowable deflection of L/240 was used to generate this table.

3) Deflections are based on positive (inward) forces only.

4) This table assumes 2 Hidden Screws with Clip Only.

5) This table assumes 1 interior clip with 2 fastener(s) and 1 end clip with 2 fastener(s).

6) The calculated dead weight of the panel is 3.05 psf and has not been considered in the tabulated loads shown above.

7) Panels must be checked independently for thermal stress and manufacturing and shipping considerations.

8) This analysis was prepared on May 3, 2017 using SpanPro Release 3, Beta 12.

# 4" KingZip 24/24 - 42"
## Live Load Analysis

KINGSPAN

"Long Term" Snow load

Stair Tower Purlin Span

Building Purlin Span

### Allowable Load for Panel

| Span | Single Span | | Double Span | | Triple Span | |
|---|---|---|---|---|---|---|
| Ft - In | Load PSF | Governing Factor | Load PSF | Governing Factor | Load PSF | Governing Factor |
| 2' - 0" | 173.48 | Deflection | 173.50 | Deflection | 173.52 | Deflection |
| 2' - 6" | 138.19 | Deflection | 138.22 | Deflection | $\boxed{138.26}$ | Deflection |
| 3' - 0" | 114.55 | Deflection | 114.61 | Deflection | 114.67 | Deflection |
| 3' - 6" | 97.58 | Deflection | 97.67 | Deflection | $\boxed{97.75}$ | Deflection |
| 4' - 0" | 84.77 | Deflection | 84.91 | Deflection | 85.02 | Deflection |
| 4' - 6" | 74.76 | Deflection | 74.95 | Deflection | 75.08 | Deflection |
| 5' - 0" | 66.69 | Deflection | 66.94 | Deflection | $\boxed{67.11}$ | Deflection |
| 5' - 6" | 60.04 | Deflection | 60.36 | Deflection | 60.56 | Deflection |
| 6' - 0" | 54.46 | Deflection | 54.85 | Deflection | 55.08 | Deflection |
| 6' - 6" | 49.70 | Deflection | 50.18 | Deflection | 50.43 | Deflection |
| 0" | 45.60 | Deflection | 46.16 | Deflection | 46.44 | Deflection |
| 6" | 42.02 | Deflection | 42.67 | Deflection | 42.97 | Deflection |
| 8' - 0" | 38.86 | Deflection | 39.61 | Deflection | 39.92 | Deflection |
| 8' - 6" | 36.06 | Deflection | 36.91 | Deflection | 37.23 | Deflection |
| 9' - 0" | 33.56 | Deflection | 34.50 | Deflection | 34.83 | Deflection |
| 9' - 6" | 31.30 | Deflection | 32.35 | Deflection | 32.67 | Deflection |
| 10' - 0" | 29.26 | Deflection | 30.41 | Deflection | 30.73 | Deflection |
| 10' - 6" | 27.41 | Deflection | 28.65 | Deflection | 28.96 | Deflection |
| 11' - 0" | 25.72 | Deflection | 27.05 | Deflection | 27.36 | Deflection |
| 11' - 6" | 24.18 | Deflection | 25.60 | Deflection | 25.89 | Deflection |
| 12' - 0" | 22.75 | Deflection | 24.26 | Deflection | 24.54 | Deflection |

NOTES:
1) Section properties, ultimate values and safety factors used to generate this table:
   - $I_c$ = 2.7370 $In^4/Ft$   $S_{bf}$ = 0.7512 $In^3/Ft$   $S_{bl}$ = 1.2834 $In^3/Ft$
   - $A_c$ = 48.45 $In^2/Ft$   $G_c$ = 217 psi (effect of creep considered)   E = 29,500,000 psi
   - $F_b$ = 24852 psi w/ Safety Factor = 2.50   $F_v$ = 17.80 psi w/ Safety Factor = 3.00
2) An allowable deflection of L/240 was used to generate this table.
3) Deflections are based on positive (inward) forces only.
4) This table assumes 2 Hidden Screws with Clip Only.
5) This table assumes 1 interior clip with 2 fastener(s) and 1 end clip with 2 fastener(s).
6) The calculated dead weight of the panel is 3.05 psf and has not been considered in the tabulated loads shown above.
7) Panels must be checked independently for thermal stress and manufacturing and shipping considerations.
8) This analysis was prepared on May 3, 2017 using SpanPro Release 3, Beta 12.

Appendix C: Metlspan Roof/Wall Information



# Load Span Tables

**The Load Span Tables below are based on Allowable Stress Design (ASD). For loads calculated based on ASCE 7-10 (LRFD), please refer to section 2.4.1 of ASCE 7-10 for the applicable load combinations using Allowable Stress Design.**

**CFR-42 ROOF SYSTEM ALLOWABLE UNIFORM LOAD CHART**

| PANEL THICKNESS | DESIGN CRITERIA | ALLOWABLE LOAD (PSF) | | | | | |
|---|---|---|---|---|---|---|---|
| | | PANEL SPAN (ft) | | | | | |
| | | 2.5' | 3.0' | 4.0' | 5.0' | 6.0' | 7.0' |
| 2" THICK 2 FASTENERS/CLIP | BENDING & SHEAR | 143.8 | 118.0 | 86.2 | 67.7 | 55.7 | 47.3 |
| | DEFLECTION (L/240) | 120.3 | 98.3 | 70.6 | 53.8 | 42.4 | 34.3 |
| | CONNECTION | 58.5 | 53.5 | 46.8 | 42.5 | 34.5 | 28.5 |
| 2.5" THICK 2 FASTENERS/CLIP | BENDING & SHEAR | 165.5 | 135.8 | 99.3 | 77.9 | 64.0 | 54.3 |
| | DEFLECTION (L/240) | 147.5 | 121.0 | 87.6 | 67.4 | 53.7 | 43.9 |
| | CONNECTION | 65.0 | 58.5 | 50.3 | 45.1 | 36.6 | 30.4 |
| 3" THICK 2 FASTENERS/CLIP | BENDING & SHEAR | 184.9 | 152.0 | 111.2 | 87.2 | 71.5 | 60.6 |
| | DEFLECTION (L/240) | 171.5 | 141.1 | 102.8 | 79.6 | 63.9 | 52.7 |
| | CONNECTION | 71.3 | 63.5 | 53.6 | 47.6 | 38.7 | 32.2 |
| 4" THICK 2 FASTENERS/CLIP | BENDING & SHEAR | 196.2 | 161.6 | 118.6 | 93.0 | 76.2 | 64.5 |
| | DEFLECTION (L/240) | 210.0 | 173.4 | 127.3 | 99.4 | 80.6 | 67.2 |
| | CONNECTION | 84.0 | 73.4 | 59.9 | 51.8 | 42.4 | 35.7 |
| 5" THICK 3 FASTENERS/CLIP | BENDING & SHEAR | 223.1 | 184.2 | 135.6 | 106.5 | 87.3 | 73.8 |
| | DEFLECTION (L/240) | 235.4 | 194.9 | 143.8 | 112.9 | 92.2 | 77.3 |
| | CONNECTION | 86.5 | 75.5 | 61.8 | 53.4 | 44.4 | 37.9 |
| 6" THICK 3 FASTENERS/CLIP | BENDING & SHEAR | 247.2 | 204.5 | 151.0 | 118.9 | 97.6 | 82.5 |
| | DEFLECTION (L/240) | 247.8 | 205.4 | 152.2 | 120.1 | 98.5 | 82.9 |

Building

Stair tower purling spacing at end is 1m (3.28ft). Therefore, considering load table for 4ft is conservative.

Stair Tower

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | CONNECTION | 88.8 | 77.6 | 63.5 | 55.0 | 46.3 | 40.0 |

- *Based on CFR-42 panel with 24 ga. exterior face (min Fy = 50 ksi) and 26 ga. interior face (min Fy = 33 ksi).*
- *Based on attachment at interior supports with CFR panel clip and (2 or 3 as shown above) 1/4"-14 Self-Drilling Tek 3 screws in min. 14 gage steel or (2) 1/4"-14 Self-Drilling Tek 3 screws in min. 12 gage steel. Two fasteners per clip are required at end supports. In lieu of self-drilling screws, self-tapping screws may be used.*
- *Allowable positive load is the lowest value of panel bending strength, shear strength & deflection limit.*
- *Allowable suction load is the lowest value of panel bending strength, shear strength, deflection limit and connection strength for each fastener pattern.*
- *Connection loads may be increased with Fablok. Consult Metl-Span for additional loads.*
- *The loads based on panel stress and deflection design criteria are derived from ASTM E-72 structural testing. The allowable loads are calculated with a factor of safety of 2.5 and 3.0 for bending and shear stresses, respectively, and deflection limitation of L/240 and with a factor of safety of 2.0 for connection strength.*
- *The clip fastener capacity was determined from manufacturer fastener pullout data and the allowable loads are calculated with a factor of safety of 3.0.*
- *The structural capacity of the purlins are not considered and must be examined independently.*
- *Multiple spans are based on 3 or more spans conditions.*
- *For allowable snow loads please consult factory.*



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|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 3″ | 109.4 | 85.6 | 70.0 | 58.9 | 50.8 | 44.6 | 39.7 | 35.4 | 30.7 |
| | | 4″ | 112.3 | 88.3 | 72.3 | 61.0 | 52.6 | 46.1 | 41.0 | 36.9 | 33.5 |
| 36″ wide | FP1 | 2″ | 54.5 | 42.6 | 34.8 | 29.3 | 25.3 | 22.3 | 19.9 | 18.0 | 16.4 |
| | | 2.5″ | 56.5 | 44.1 | 36.0 | 30.3 | 26.2 | 23.0 | 20.5 | 18.5 | 16.8 |
| | | 3″ | 58.2 | 45.6 | 37.2 | 31.4 | 27.0 | 23.7 | 21.1 | 19.0 | 17.3 |
| | | 4″ | 61.4 | 48.3 | 39.5 | 33.3 | 28.7 | 25.2 | 22.4 | 20.2 | 18.3 |
| | FP2 | 2″ | 65.6 | 51.2 | 41.8 | 35.3 | 30.5 | 26.8 | 23.9 | 21.6 | 18.7 |
| | | 2.5″ | 71.1 | 55.5 | 45.3 | 38.2 | 33.0 | 29.0 | 25.8 | 23.3 | 21.2 |
| | | 3″ | 76.4 | 59.8 | 48.9 | 41.2 | 35.5 | 31.1 | 27.7 | 25.0 | 22.7 |
| | | 4″ | 86.8 | 68.2 | 55.9 | 47.1 | 40.6 | 35.6 | 31.7 | 28.5 | 25.9 |
| | FP3 | 2″ | 87.9 | 68.6 | 56.0 | 45.8 | 37.6 | 31.2 | 26.1 | 22.0 | 18.7 |
| | | 2.5″ | 89.6 | 70.0 | 57.2 | 48.2 | 41.6 | 36.5 | 32.6 | 29.1 | 25.0 |
| | | 3″ | 91.1 | 71.3 | 58.3 | 49.1 | 42.3 | 37.2 | 33.1 | 29.8 | 27.1 |
| | | 4″ | 93.6 | 73.6 | 60.3 | 50.8 | 43.8 | 38.4 | 34.2 | 30.8 | 27.9 |

*Notes:*

1. *The Load Span Table above is based on Allowable Stress Design (ASD). for loads calculated based on ASCE 7-10 (LRFD), please refer to section 2.4.1 of ASCE 7-10 for the applicable load combinations using Allowable Stress Design.*
2. *Based on CF-panel with 24 ga. Striated exterior and 26 ga. Striated interior face (min $F_y$ = 33ksi).*
3. *Fastener pattern FP1 is based on CF panel clips fastened to min. 14 ga. steel. Fastener options will be (2) ¼″-14 SDS type 3, (2) ¼″-14 Self-Tapping, (2) ¼″-14 Type 5 SDS, (2) ¼″-20 Type 5 SDS, or (2)¼″-28 Type 5 SDS. Fastener selection will be based on fastener pullout capacity from support steel members.*
4. *For CF-30, FP2 is based on FP1 along with (1) blind rivet at 10" o.c. from female panel sidelap. For CF-36, FP2 is based on FP1 along with (1) blind rivet at 12" o.c. from female panel sidelap.*
5. *Fro CF-30, FP3 is based on FP1 along with (2) blind rivets at 10" o.c. from female panel sidelap. For CF-36, FP3 is based on FP1 along with (2) blind rivets at 12" o.c. from female panel sidelap.*
6. *Allowable loads based on panel stress, connection strength and deflection design criteria are derived from ASTM E72 and E1592 structural testing.*
7. *The allowable inward or outward loads is the smallest load calculated with a factor of safety of 2.5 for bending stress, 3.0 for shear stresses, 2.0 for connection and deflection limitation of L/180.*
8. *The structural capacity of the supports are not considered and must be examined independently.*

## CF-42 Striated Wall Panels

## 24 Ga. Exterior / 26 Ga. Interior Facings

## Allowable Connection Load[1,5,6,7,8] (psf) for Two or More Equal Spans

| Panel Type[2] | Design Criteria[3,4] | Support Span (ft) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 5′ | 6′ | 7′ | 8′ | 9′ | 10′ | 11′ | 12′ |
| 2″ Thick | Connection FP1 | 31.6 | 25.8 | 21.8 | 18.8 | 16.6 | 14.8 | 13.3 | 12.2 |
| | Connection FP2 | 46.3 | 37.8 | 31.9 | 27.5 | 24.2 | 21.6 | 19.5 | 17.8 |
| | Connection FP3 | 58.4 | 47.7 | 40.2 | 34.8 | 30.6 | 26.1 | 22.0 | 18.7 |
| 2.5″ Thick | Connection FP1 | 33.5 | 27.3 | 23.0 | 19.9 | 17.4 | 15.5 | 14.0 | 12.8 |
| | Connection FP2 | 47.7 | 38.9 | 32.8 | 28.3 | 24.9 | 22.2 | 20.0 | 18.2 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Connection FP3 | 60.3 | 49.2 | 41.5 | 35.8 | 31.4 | 28.0 | 25.3 | 23.0 |
| 3" Thick | Connection FP1 | 35.3 | 28.8 | 24.3 | 20.9 | 18.4 | 16.3 | 14.7 | 13.4 |
| | Connection FP2 | 49.0 | 40.0 | 33.7 | 29.1 | 25.5 | 22.7 | 20.5 | 18.6 |
| | Connection FP3 | 62.1 | 50.7 | 42.7 | 36.9 | 32.4 | 28.8 | 26.0 | 23.6 |
| | Connection FP4 | 71.4 | 58.3 | 49.1 | 42.4 | 37.2 | 33.1 | 29.8 | 27.2 |
| 4" Thick | Connection FP1 | 43.4 | 35.6 | 30.0 | 25.9 | 22.7 | 20.2 | 18.1 | 16.5 |
| | Connection FP2 | 63.6 | 52.1 | 44.0 | 37.9 | 33.2 | 29.6 | 26.6 | 24.2 |
| | Connection FP3 | 75.1 | 61.5 | 51.8 | 44.7 | 39.2 | 34.9 | 31.4 | 28.5 |
| | Connection FP4 | 80.4 | 65.8 | 55.5 | 47.9 | 42.0 | 37.4 | 33.6 | 30.5 |
| | Connection FP5 | 84.1 | 68.9 | 58.1 | 50.1 | 43.9 | 39.1 | 35.2 | 32.0 |
| | Connection FP9 | 90.4 | 74.1 | 62.5 | 53.8 | 47.2 | 42.0 | 37.8 | 34.3 |
| | Connection FP10 | 94.6 | 77.5 | 65.3 | 56.3 | 49.4 | 44.0 | 39.5 | 35.9 |

*Notes:*

1. *The Load Span Table above is based on Allowable Stress Design (ASD). For loads calculated based on ASCE 7-10 (LFD), please refer to section 2.4.1 of ASCE 7-10 for the applicable load combinations using Allowable Stress Design.*
2. *Based on CF-42 panel with 24 ga. Striated exterior and 26 ga. Light Mesa interior face (min $F_y$ = 33kis).*
3. *Fastener pattern FP1 is based on CF panel clips fastened to min. 14 ga. steel. Fastener options will be (2) ¼"-14 SDS Type 3, (2) ¼"-14 Self-Tapping, (2) ¼"-14 Type 5 SDS, (2) ¼"-20 Type 5 SDS, or (2) ¼"-28 Type 5 SDS. Fastener selection will be based on fastener pullout capacity from support steel members.*
4. *The fastener patterns FP2, FP3, FP4, and FP5, include FP1 plus 1, 2, 3, and 4 blind rivets, respectively, at supports per panel width. Blind rivet spacing is 10.5" o.c. from female edge of panel seam.*
5. *The through fasteners (FP9) are as follows: Panels fastened to min. 14 ga. steel supports with (4) ¼"-14 SDS Type 3, (4) ¼"-14 Self-Tapping, (4) ¼"-14 Type 5 SDS, (4)¼"-20 Type 5 SDS, or (4) ¼"-28 Type 5 SDS with nominal 5/8" diameter neoprene bonded washers spaced at 8.4" o.c. Fastener shall be of sufficient length to penetrate through the support a minimum of 3/4". Fastener selection will be based on fastener pullout capacity from support steel members.*
6. *The through fasteners (FP10) are as follows: Panels fastened to min. 14 ga. steel supports with (5) ¼"-14 SDS Type 3, (5) ¼"-14 Self-Tapping, (5) ¼"-14 Type 5 SDS, (5) ¼"-20 Type 5 SDS, or (5) ¼"-28 Type 5 SDS with nominal 5/8" diameter neoprene bonded washers spaced at 8.4" o.c. Fastener shall be of sufficient length to penetrate through the support a minimum of 3/4". Fastener selection will be based on fastener pullout capacity from support steel members.*
7. *Allowable positive or suction load is the lowest value of panel bending strength, shear strength, deflection limit and connection strength for each fastener pattern.*
8. *Allowable loads based on panel stress and deflection design criteria are derived from ASTM E72 structural testing and calculated with factor of safety of 2.5 for bending stress, 3.0 for shear stresses and deflection limitation of L/180.*
9. The panel connection strength was determined from ASTM E1592 testing and the allowable loads are calculated with factor of safety of 2.
10. The structural capacity of the purlins are not considered and must be examined independently.

## CF-42 Striated Wall Panel[7]

## 24 Ga. Exterior / 26 Ga. Interior Facings

## Allowable Positive Load[1,4,5,6] (psf) for Two or More Equal Spans

| Panel Type[2] | Design Criteria[3] | Support Span (ft) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 5' | 6' | 7' | 8' | 9' | 10' | 11' | 12' |
| 2" Thick | Bending & Shear | 71.2 | 58.6 | 49.8 | 43.1 | 37.9 | 33.9 | 27.5 | 22.5 |
| | Deflection (L/180) | 71.8 | 56.6 | 45.8 | 37.6 | 31.2 | 26.1 | 22.0 | 18.7 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2.5" Thick | Bending & Shear | 81.0 | 66.6 | 56.5 | 49.0 | 43.2 | 38.5 | 33.9 | 27.4 |
| | Deflection (L/180) | 87.5 | 69.6 | 56.9 | 47.5 | 40.0 | 34.0 | 29.1 | 25.0 |
| 3" Thick | Bending & Shear | 90.1 | 73.9 | 62.6 | 54.3 | 47.9 | 42.9 | 38.6 | 32.5 |
| | Deflection (L/180) | 100.6 | 80.6 | 66.4 | 55.8 | 47.5 | 40.9 | 35.4 | 30.7 |
| 4" Thick | Bending & Shear | 95.5 | 78.3 | 66.2 | 57.3 | 50.5 | 45.1 | 40.8 | 37.2 |
| | Deflection (L/180) | 118.5 | 96.0 | 79.8 | 67.7 | 58.3 | 508 | 44.6 | 39.6 |

*Notes:*

1. *The Load Span Table above is based on Allowable Stress Design (ASD). For loads calculated based on ASCE 7-10 (LRFD), please refer to section 2.4.1 of ASCE 7-10 for the applicable load combinations using Allowable Stress Design.*
2. *Based on CF-panel with 24 ga. Striated exterior and 26 ga. Light Mesa interior face (min Fy = 33 ksi).*
3. *Refer to the allowable connection load chart, for suction loads.*
4. *Allowable positive or suction load is the lowest value of panel bending strength, shear strength, deflection limit and connection strength for each fastener pattern.*
5. *Allowable loads based on panel stress and deflection design criteria are derived from ASTM E72 structural testing and calculated with factor of safety of 2.5 for bending stress, 3.0 for shear stresses and deflection limitation of L/180.*
6. *The structural capacity of the purlins are not considered and must be examined independently.*
7. *Consult Metl-Span for recommendtions on panel profile and gage suitable for thermal stresses.*



***Toll Free:*** 877.585.9969
***Phone:*** 972.221.6656
***Fax:*** 972.420.9382
1720 Lakepointe Dr, Ste. #101
Lewisville, TX 75057

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Connection TSFP1[8] | 62.3 | 51.2 | 43.3 | 37.4 | 32.8 | 29.2 | 26.3 | 23.8 | 21.8 | 20.1 | 18.6 | 17.3 |
| | Connection TSFP2[9] | 83.1 | 68.3 | 57.8 | 49.9 | 43.8 | 39.0 | 35.0 | 31.8 | 29.1 | 26.8 | 24.8 | 23.1 |

1. The Load Span Table above is based on Allowable Stress Design (ASD). For loads calculated based on ASCE 7-10 (LRFD), please refer to section 2.4.1 of ASCE 7-10 for the applicable load combinations using Allowable Stress Design.
2. Based on TS panel with 24 Ga. exterior & 26 Ga. interior face (min Fy = 33 ksi) for 2 or more spans condition.
3. Allowable positive or inward load is the lowest value of panel bending and shear strength or deflection limit.
4. Allowable suction or outward load is the lowest value of panel bending and shear strength, deflection limit and connection strength for each fastener pattern. The numbers have been reduced to reflect the lowest value.
5. Loads based on panel stress, deflection and connection design criteria are derived from ASTM E-72 testing.
6. Allowable loads are calculated with a factor of safety of 2.5 for bending, 3.0 for shear stresses and 2.0 for connection, and deflection limitation of L/240.
7. The through fasteners are as follows: Panels fastened to min 16 ga. steel supports with 1/4"-Type 3 or Type 5 SDS, 1/4"-14 Self-Tapping, 1/4"-20 Type 5 SDS, or 1/4"-28 Type 5 SDS with nominal 5/8" diameter neoprene bonded washers. Fasteners shall be sufficient length to penetrate through the support a minimum of 3 full pitches of thread. Other than noted, fastener selection will be based on fastener pullout capacity from support steel members.
8. TSFP1 (3 Through Fasteners): End Support (3"-18"-18"-3") & Intermediate (7"-14"-14"-7").
9. TSFP2 (4 Through Fasteners): End Supports (3"-12"-12"-12"-3") & Intermediate (5 1/4"-10 1/2"-10 1/2"-10 1/2"-5 1/4").
10. The structural capacity of the girts are not considered and must be examined independently.

**Metl-Span ThermalSafe Through Fastened Wall Panels**
**24 Ga. Exterior / 24 Ga. Interior Facings**
**Allowable Loads[1,2,6,10] (psf) for Two or More Spans Condition**

| **TS Panel** | **Design Criteria[5,7]** | **Panel Span (ft)** | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **5'** | **6'** | **7'** | **8'** | **9'** | **10'** | **11'** | **12'** | **13'** | **14'** | **15'** | **16'** |
| 4" Thick | Bending & Shear[3,4] | 72.0 | 59.1 | 50.0 | 43.3 | 38.1 | 34.1 | 30.8 | 28.1 | 25.7 | 23.8 | 22.0 | 20.6 |
| | Deflection (L/240)[3,4] | 119.4 | 96.3 | 79.9 | 67.5 | 57.9 | 50.3 | 44.1 | 38.8 | 34.3 | 30.4 | 27.1 | 24.3 |
| | Connection TSFP1[8] | 60.7 | 49.6 | 41.8 | 36.1 | 31.6 | 28.1 | 25.3 | 23.0 | 21.1 | 19.5 | 18.1 | 16.9 |
| | Connection TSFP2[9] | 72.0 | 59.1 | 50.0 | 43.3 | 38.1 | 34.1 | 30.8 | 28.1 | 25.7 | 23.8 | 22.0 | 20.6 |
| 5" Thick | Bending & Shear[3,4] | 85.7 | 70.2 | 59.4 | 51.4 | 45.2 | 40.4 | 36.5 | 33.3 | 30.6 | 28.2 | 26.2 | 24.4 |
| | Deflection (L/240)[3,4] | 144.7 | 117.4 | 97.9 | 83.2 | 71.8 | 62.6 | 55.2 | 49.0 | 43.8 | 39.3 | 35.3 | 31.8 |
| | Connection TSFP1[8] | 61.4 | 50.3 | 42.5 | 36.6 | 32.1 | 28.6 | 25.7 | 23.3 | 21.4 | 19.7 | 18.3 | 17.0 |
| | Connection TSFP2[9] | 81.9 | 67.1 | 56.6 | 48.8 | 42.8 | 3.8.1 | 34.3 | 31.1 | 28.5 | 26.3 | 24.4 | 22.7 |
| 6" Thick | Bending & Shear[3,4] | 97.3 | 79.8 | 67.4 | 58.3 | 51.3 | 45.8 | 41.3 | 37.7 | 34.6 | 32.0 | 29.7 | 27.7 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Deflection (L/240)[3,4] | 166.3 | 135.5 | 113.4 | 96.8 | 83.9 | 73.5 | 65.0 | 58.0 | 52.1 | 47.0 | 42.6 | 38.7 |
| | Connection TSFP1[8] | 62.0 | 50.9 | 43.0 | 37.1 | 32.6 | 29.0 | 26.0 | 23.6 | 21.6 | 19.9 | 18.5 | 17.2 |
| | Connection TSFP2[9] | 82.7 | 57.9 | 57.3 | 49.5 | 43.4 | 38.6 | 34.7 | 31.5 | 28.9 | 26.6 | 24.6 | 23.0 |
| 7" Thick | Bending & Shear[3,4] | 114.5 | 93.9 | 79.4 | 68.6 | 60.3 | 53.8 | 48.6 | 44.3 | 40.6 | 37.6 | 34.9 | 32.6 |
| | Deflection (L/240)[3,4] | 195.9 | 160.0 | 134.3 | 114.9 | 99.7 | 87.6 | 77.7 | 69.5 | 62.5 | 56.6 | 51.5 | 47.0 |
| | Connection TSFP1[8] | 62.3 | 51.2 | 43.3 | 37.4 | 32.8 | 29.2 | 26.3 | 23.8 | 21.8 | 20.1 | 18.6 | 17.3 |
| | Connection TSFP2[9] | 83.1 | 68.3 | 57.8 | 49.9 | 43.8 | 39.0 | 35.0 | 31.8 | 29.1 | 26.8 | 24.8 | 23.1 |
| 8" Thick | Bending & Shear[3,4] | 131.7 | 108.1 | 91.4 | 79.0 | 69.5 | 62.0 | 55.9 | 50.9 | 46.7 | 43.2 | 40.1 | 37.5 |
| | Deflection (L/240)[3,4] | 225.5 | 184.6 | 155.2 | 133.0 | 115.7 | 101.9 | 90.5 | 81.1 | 73.1 | 66.3 | 60.4 | 55.3 |
| | Connection TSFP1[8] | 62.6 | 51.5 | 43.6 | 37.7 | 33.1 | 29.4 | 26.5 | 24.0 | 22.0 | 20.2 | 18.7 | 17.5 |
| | Connection TSFP2[9] | 83.4 | 68.7 | 58.1 | 50.2 | 44.1 | 39.2 | 35.3 | 32.0 | 29.3 | 27.0 | 25.0 | 23.3 |

1. The Load Span Table above is based on Allowable Stress Design (ASD). For loads calculated based on ASCE 7-10 (LRFD), please refer to section 2.4.1 of ASCE 7-10 for the applicable load combinations using Allowable Stress Design.
2. Based on TS panel with 24 Ga. exterior & 24 Ga. interior face (min Fy = 33 ksi) for 2 or more spans condition.
3. Allowable positive or inward load is the lowest value of panel bending and shear strength or deflection limit.
4. Allowable suction or outward load is the lowest value of panel bending and shear strength, deflection limit and connection strength for each fastener pattern. The numbers have been reduced to reflect the lowest value.
5. Loads based on panel stress, deflection and connection design criteria are derived from ASTM E-72 testing.
6. Allowable loads are calculated with a factor of safety of 2.5 for bending, 3.0 for shear stresses and 2.0 for connection, and deflection limitation of L/240.
7. The through fasteners are as follows: Panels fastened to min 16 ga. steel supports with 1/4"-Type 3 or Type 5 SDS, 1/4"-14 Self-Tapping, 1/4"-20 Type 5 SDS, or 1/4"-28 Type 5 SDS with nominal 5/8" diameter neoprene bonded washers. Fasteners shall be sufficient length to penetrate through the support a minimum of 3 full pitches of thread. Other than noted, fastener selection will be based on fastener pullout capacity from support steel members.
8. TSFP1 (3 Through Fasteners): End Support (3"-18"-18"-3") & Intermediate (7"-14"-14"-7").
9. TSFP2 (4 Through Fasteners): End Supports (3"-12"-12"-12"-3") & Intermediate (5 1/4"-10 1/2"-10 1/2"-5 1/4").
10. The structural capacity of the girts are not considered and must be examined independently.

**Allowable Span for Inward or Outward Load of 10 psf**

| Thickness (in) | Design Criteria | | | Allowable Span (ft) |
|---|---|---|---|---|
| | Bending Stress | Shear Stress | Deflection Limit | |
| 4" | 21.7 | 40.0 | 19.8 | 19.8 |
| 5" | 24.3 | 46.9 | 22.9 | 22.9 |

FLUOR

**Fw: Product Info**
Lyndon Trombley to: Kharla Valerio 05/04/2017 10:32 AM

From: Lyndon Trombley/CA/FD/FluorCorp
To: Kharla Valerio/CA/FD/FluorCorp@FluorCorp,

Kharla,
FYI

Sincerely,
**Lyndon Trombley, P. Eng (AB)** I **FLUOR** I CSA - Williams PDH Project I Calgary, AB I
Lyndon.Trombley@fluor.com I 403.537.5134 (IODC 8-88-2085)

*Measure Twice, Cut Once*

----- Forwarded by Lyndon Trombley/CA/FD/FluorCorp on 05/04/2017 10:32 AM -----

From: Larry Dimalanta <lcdimalanta@metlspan.com>
To: "'Lyndon.Trombley@fluor.com'" <Lyndon.Trombley@fluor.com>, calculations <calculations@metlspan.com>,
Cc: "Michael J. Lemen" <MJLemen@metlspan.com>, Geoff Seale <Geoff.Seale@MetlSpan.com>
Date: 05/04/2017 08:49 AM
Subject: RE: Product Info

---

Lyndon,

The answers to your questions are as follows:

1. In general, the roof panels are governed by faster capacity for suction. ASD load combinations allow Wind loads to be resisted by 0.6 Dead Load. We do not use the dead load of the panels to resist uplift in our calculations. You may increase the connection capacity by including 0.6 panel Dead Load to the table value.
2. For Roof panel, allowable Gravity Loads values in the chart have accounted for panel weight.
3. The load/span tables are a combination of testing and analysis. It would be cost prohibitive to test every entry in our charts. Tests are performed to establish the section properties of the panel. Calculations are used from the tested section properties to determine maximum loads for various spans. Several tests are done based on minimum and maximum thicknesses as well as minimum and maximum spans to confirm that the calculated values match test results.

I trust this is sufficient for your needs. Feel free to contact me with any questions, comments, or concerns that you may have.

Regards,

Lorenzo (Larry) Dimalanta, PE[1], PEng[2], MCP
Metl-Span | Design and Compliance Engineer
[1] Licensed in AZ, CA, CO, DC, FL, GA, GU, IA, IL, IN, LA, MI, MS, NC, NY, OK, TX, and VA
[2] Licensed in AB, BC, ON, SK, and YK
P: 972.538.9040
F: 972.420.6790

**From:** Geoff Seale
**Sent:** Wednesday, May 03, 2017 5:28 PM

**To:** Rahim Hakimi; Larry Dimalanta
**Cc:** Michael J. Lemen
**Subject:** FW: Product Info

Good Afternoon
Can you review the email string and provide an answer to Lyndon please

Thanks

Geoff Seale
Sales & Marketing Manager - Western Canada

METL SPAN CANADA

Cell: 587.987.2150
Geoff.Seale@metlspan.com

**From:** Lyndon.Trombley@fluor.com [mailto:Lyndon.Trombley@fluor.com]
**Sent:** Wednesday, May 3, 2017 2:11 PM
**To:** Geoff Seale
**Cc:** Kharla.Valerio@fluor.com; Michael J. Lemen
**Subject:** RE: Product Info

Geoff,
I apologize I might not have asked the questions in the most clear way. By asking if the dead load of the panel is included or not I need to ask the question a bit differently.

Please correct me if I misinterpret the information -

-for gravity load (like snow) are the loads given including the dead weight of the panel, and the allowable external imposed load would be the tabulated value MINUS the panel self weight. Accordingly, for a suction load would one be able to take advantage of the self-weight by takeing the tabulated value and adding back the self weight of the panel?

Sincerely,
**Lyndon Trombley, P. Eng (AB)** l **FLUOR** l CSA - Williams PDH Project l Calgary, AB l

Lyndon.Trombley@fluor.com l 403.537.5134 (IODC 8-88-2085)

*Measure Twice, Cut Once*

From: Geoff Seale <Geoff.Seale@MetlSpan.com>
To: "Lyndon.Trombley@fluor.com" <Lyndon.Trombley@fluor.com>,
Cc: "Michael J. Lemen" <MJLemen@metlspan.com>, "Kharla.Valerio@fluor.com" <Kharla.Valerio@fluor.com>

Date: 05/02/2017 05:31 PM
Subject: RE: Product Info

Hello Lyndon
Please send over any questions you may have and we will do our best to answer in a timely fashion.
We certainly can produce a 6” 7” and 8” fire rated panel which as you said would be our Thermalsafe panels
I believe the answers to both of the last questions you have is yes however Mike and I will look into it and get back to you if there is any difference

Thanks

Geoff Seale
Sales & Marketing Manager - Western Canada

METL SPAN CANADA

Cell: 587.987.2150
Geoff.Seale@metlspan.com

**From:** Lyndon.Trombley@fluor.com [mailto:Lyndon.Trombley@fluor.com]
**Sent:** Tuesday, May 2, 2017 3:39 PM
**To:** Geoff Seale
**Cc:** Michael J. Lemen; Kharla.Valerio@fluor.com
**Subject:** Re: Product Info

Geoff,
I may have other questions as I get deeper into it, we are at the design stage on an industrial building and have multiple panel manufacturer's approved by the owner's standard already so digesting a fair bit of information.

Thanks for getting back to me so fast. The inquiry I made on your "contact us" web page should be disregarded as I believe we need a small amount of 6" fire-rated panel which would be the thermalsafe panels and not striated.

My general questions (so far):

-for roof panels subject to suction, are the allowable loads (which seem to be governed by fastener capacity generally) presented including the self-weight of the panel?
-are the load/span tables based on analysis or testing (to ASTM E72) or a combination?

Sincerely,
**Lyndon Trombley, P. Eng (AB)** I **FLUOR** I CSA - Williams PDH Project I Calgary, AB I Lyndon.Trombley@fluor.com

l 403.537.5134 (IODC 8-88-2085)

**Measure Twice, Cut Once**

From: Geoff Seale <Geoff.Seale@MetlSpan.com>
To: "lyndon.trombley@fluor.com" <lyndon.trombley@fluor.com>,
Cc: "Michael J. Lemen" <MJLemen@metlspan.com>
Date: 05/02/2017 10:15 AM
Subject: Product Info

---

Good morning Lyndon
I was forwarded your request from our head office. I take care of Western Canada so feel free at any time to call or email me and I will do my best to help you out.
In response to your question with regard to the 6" striated panel, we will only produce this panel in a 4" thickness.
We will make a Mesa, Fluted and Santa Fe in a 6" however.
If you have a certain requirement for the profile or the thickness I would be more than happy to discuss this with you and I would assume that economics are also a consideration.
If this product is going into Alaska I have also copied Mike Lemen on this email as he takes care of the western USA, Alaska and Hawaii.
Please feel free to contact us at any time

Thanks


Geoff Seale
Sales & Marketing Manager - Western Canada

METL SPAN CANADA

Cell: 587.987.2150
Geoff.Seale@metlspan.com